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Functional characterization of the disease-associated *CCL2* rs1024611G-rs13900T haplotype: The role of the RNA-binding protein HuR

Feroz Akhtar, Joselin Hernandez Ruiz, Ya-Guang Liu, Roy G Resendez, Denis Feliers, Liza D Morales, Alvaro Diaz-Badillo, Donna M Lehman, Rector Arya, Juan Carlos Lopez-Alvarenga, John Blangero, Ravindranath Duggirala, Srinivas Mummidi

Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States • Utah Center for Genetic Discovery, Department of Human Genetics, University of Utah, Salt Lake City, United States • Department of Pathology, School of Medicine, University of Texas Health San Antonio, San Antonio, United States • Department of Medicine, School of Medicine, University of Texas Health San Antonio, San Antonio, United States • South Texas Diabetes and Obesity Institute, Department of Genetics, School of Medicine, University of Texas Rio Grande Valley, Brownsville, United States • Department of Population Health and Biostatistics, School of Medicine, University of Texas Rio Grande Valley, Harlingen, United States

eLife Assessment

CCL2 is a chemokine with immune cell chemoattractant properties, and it appears to play a role in several chronic inflammatory diseases. The RNA-binding protein HuR controls the stability and translation of CCL2 mRNA. This paper presents **convincing** evidence that a relatively common genetic variant tied to several disease phenotypes affects the interaction between the mRNA of CCL2 and the RNA-binding protein HuR. While the experiments cannot definitively distinguish between effects on RNA transcription and stability, CCL2 is thought to be relevant for leukocyte migration in various conditions, including chronic inflammation and cancer, and the study presents **important** findings that may be relevant to a broad audience.

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Abstract

CC-chemokine ligand 2 (CCL2) is involved in the pathogenesis of several diseases associated with monocyte/macrophage recruitment, such as HIV-associated neurocognitive disorder (HAND), tuberculosis, and atherosclerosis. The rs1024611 (alleles: A>G; G is the risk allele) polymorphism in the *CCL2 cis*-regulatory region is associated with increased CCL2 expression in vitro and ex vivo, leukocyte mobilization in vivo, and deleterious disease outcomes. However, the molecular basis for the rs1024611-associated differential CCL2 expression remains poorly characterized. It is conceivable that genetic variant(s) in linkage disequilibrium (LD) with rs1024611 could mediate such effects. Previously, we used rs13900 (alleles: C>T) in the *CCL2* 3’ untranslated region (3’ UTR) that is in perfect LD with rs1024611 to demonstrate allelic expression imbalance (AEI) of *CCL2* in heterozygous individuals. Here we tested the hypothesis that the rs13900 could modulate *CCL2* expression by altering mRNA turnover and/or translatability. The rs13900 T allele conferred greater stability to the *CCL2* transcript when compared to the rs13900 C allele. The rs13900 T allele also had increased binding to Human Antigen R (HuR), an RNA-binding protein, in vitro and ex vivo. The rs13900 alleles imparted differential activity to reporter vectors and influenced the translatability of the reporter transcript. We further demonstrated the role of HuR in mediating allele-specific effects on CCL2 expression in overexpression and silencing studies. Our studies suggest that the differential interactions of HuR with rs13900 could modulate CCL2 expression and could in part explain the interindividual differences in CCL2-mediated disease susceptibility.

Introduction

Identification of functional and/or causal genetic variants continues to pose a significant challenge in the post-genome-wide association studies (post-GWAS) era (MacArthur et al., 2017; Tam et al., 2019; Visscher et al., 2017). While emphasis has been placed on polymorphisms that map to *cis*-elements in enhancers and promoters, genetic variants that disrupt cis-elements in RNA binding protein (RBP) motifs in 3' UTR have received much less attention. For many genes, the interactions of their 3' UTRs with specific stabilizing and destabilizing RBPs play a critical role in modulating post-transcriptional events such as mRNA turnover and translatability (Mayr, 2019; Schwerk and Savan, 2015). The lack of a mechanistic understanding of how SNPs localizing to RBP motifs could impact gene expression impedes a greater understanding of the variability in disease susceptibility and outcomes.

Post-transcriptional mechanisms are thought to play a crucial role in the initiation and resolution of the inflammatory response (Anderson, 2010; Yoshinaga and Takeuchi, 2019). Such regulation can profoundly affect gene expression levels; modest changes in mRNA stability can lead to significant effects on mRNA and protein abundance (Buccitelli and Selbach, 2020; Ross, 1995). Notably, a genome-scale study using mouse dendritic cells demonstrated that post-transcriptional mRNA degradation was a salient feature of inflammatory and immune signaling genes, as well as targets of NF-kappaB signaling following lipopolysaccharide (LPS) stimulation (Rabani et al., 2011). In addition, polymorphisms in non-coding regions such as the 3' UTR can have a significant impact on mRNA stability and translatability. For example, a genome-wide study of variation in gene-specific mRNA decays in lymphoblastoid cell lines across individuals found about 195 genetic variants that are specifically associated with variation in mRNA decay rates, called "rdQTLs" (RNA Decay Quantitative Trait Loci) (Pai et al., 2012). The authors estimated that 35% of the most significant eQTL (Expression Quantitative Trait Loci) single nucleotide polymorphisms (SNPs) are associated with decay rates (Pai et al., 2012). In another study, Duan *et al.* reported that ~37% of gene expression differences among individuals may be attributed to RNA half-life differences (Duan et al., 2013). Farh et al., reported that the 3' UTRs are highly enriched for eQTL candidate causal SNPs (>1,500) relative to other transcribed SNPs (Farh et al., 2015). However, the molecular mechanisms by which these polymorphisms alter mRNA stability or translatability are poorly understood.

CCL2 is a potent monocyte chemoattractant produced by various cell types, either constitutively or following activation. CCL2 expression can be regulated by inflammatory molecules (e.g., IL-1, TNFα, LPS, and IFNγ) and growth factors (e.g., PDGF). While the cells of monocyte-macrophage lineage are a major source of CCL2 (Yoshimura et al., 1989), other cell types, such as fibroblasts, astrocytes, epithelial, and endothelial cells, also are an important source (Deshmane et al., 2009). CCL2 mediates recruitment of monocytes, memory T-cells, and dendritic cells to the site of inflammation (Gschwandtner, Derler and Midwood, 2019; Melgarejo et al., 2009) and there is substantial evidence implicating CCL2 as a key mediator in macrophage-mediated diseases. Rovin et al. described a SNP in the 5'-regulatory region of *CCL2* annotated as rs1024611 (dbSNP database; originally designated as – 2518A>G (Rovin, Lu and Saxena, 1999) or –2578A>G (Gonzalez et al., 2002)) that was associated with increased plasma CCL2 expression (Rovin, Lu and Saxena, 1999). This SNP is associated with increased serum CCL2 levels, enhanced macrophage recruitment to tissues, and progression to HIV-associated dementia (Gonzalez et al., 2002). Other studies showed that the rs1024611 G allele is associated with increased CCL2 levels in the plasma, urine, and cerebrospinal fluid in health and disease, and in tissues such as liver and skin (Cho et al., 2004; Fenoglio et al., 2004; Joven et al., 2006; Letendre et al., 2004; McDermott et al., 2005). The rs1024611 polymorphism has been associated with several diseases, including myocardial infarction (McDermott et al., 2005), carotid atherosclerosis (Alonso-Villaverde et al., 2004), pulmonary tuberculosis (Flores-Villanueva et al., 2005), severe acute pancreatitis (Cavestro et al., 2010), lupus nephritis (Tucci et al., 2004), asthma susceptibility and severity (Szalai et al., 2001), Crohn's disease (CD) (Palmieri et al., 2010), Alzheimer's disease (Fenoglio et al., 2004), and infections by Japanese Encephalitis virus (Chowdhury and Khan, 2017) and SARS-CoV-1 (Tu et al., 2015). Notably, rs3091315, a GWAS risk variant for CD (Franke et al., 2010) and inflammatory bowel disease (IBD) (Liu et al., 2015) is in a strong linkage disequilibrium (LD) with both rs1024611 (D'=1.0, r^2=0.98) and rs13900 (D'=1.0, r^2=0.98) in the CEU population. Given the importance of disease associations with rs1024611, significant efforts have been made by other groups and us to understand the molecular basis of this differential CCL2 expression associated with this polymorphism (Gonzalez et al., 2002; Mummidi, Bonello and Ahuja, 2009; Page et al., 2011; Pham et al., 2012; Wright et al., 2008). However, these studies did not provide a mechanistic link between the rs1024611 polymorphism and CCL2 expression, giving rise to the possibility that SNPs in strong LD with rs1024611 could be mediating these effects.

To identify potential functional SNPs that could explain the variability in CCL2 expression, we developed an extensive LD map of the *CCL2* genomic locus and reported that a SNP designated as the rs13900 (NM_002982.4:c.*65=) in the *CCL2* 3' UTR is in perfect LD with rs1024611 and can serve as its proxy (Pham et al., 2012). We and others have demonstrated AEI in *CCL2* using rs13900 as a marker with the T allele showing a higher expression level relative to C allele (Johnson et al., 2008; Pham et al., 2012). In this study, we show that the differential binding of Human Antigen R (HuR), an RBP previously implicated in CCL2 expression, leads to altered stability and translatability of *CCL2* transcripts providing a mechanistic explanation for increased CCL2 expression in individuals with the rs1024611G-rs13900T haplotype and inter-individual differences in disease susceptibility associated with this haplotype.

Results

Individuals heterozygous for rs13900 show AEI of *CCL2*

We and others have previously reported a perfect linkage disequilibrium between rs1024611 in the *CCL2* cis-regulatory region and rs13900 in its 3' UTR and that rs13900 can serve as a proxy for the disease-associated rs1024611 (Hubal et al., 2010; Intemann et al., 2011; Kasztelewicz et al., 2017; Pham et al., 2012). We further showed that *CCL2* exhibits allelic expression imbalance (AEI) in heterozygous individuals, with the rs13900 T allele (alternative allele) having a higher expression than the rs13900 C allele (reference allele) (**Figure 1A**). For this study, we recruited 47 healthy unrelated individuals (18–35-year-old) who were screened for rs13900 (**Figure S1**). **Supplementary Table 1** shows the genotype and allele frequencies for rs13900 polymorphism in these recruited individuals. We found that the rs13900 C allele was at a higher frequency than the rs13900 T allele and the genotype frequencies were in line with Hardy-Weinberg equilibrium ($P \geq 0.05$).. We reconfirmed that *CCL2* exhibits AEI using data from heterozygous individuals. For this we used total RNA obtained from LPS treated PBMC as described previously (Pham et al., 2012). LPS induced *CCL2* expression in PBMCs as confirmed by qRT-PCR, with about a 4.3-fold increase at 1 h, 6.09-fold increase at 3 h, and 1.94-fold increase at 6 h (**Figure S2**). We used the 3 hour stimulation in the subsequent experiments as the peak *CCL2* expression was detected at this time point following LPS stimulation. AEI was measured by quantifying the relative amount of the two alleles i.e., alternative allele (T) to reference allele (C), measured from the chromatogram after normalization of peak intensity using PeakPeaker v.2.0 (**Figure 1B** and 1C). This strategy ensures a direct comparison between the amount of *CCL2* mRNA that is transcribed from each allele or haplotype and that each allele is equally subjected to the effects of any external factors. gDNA was utilized as a control. The boxplot illustrates a notable difference in the detected levels of the C and T allele in the gDNA and cDNA with a higher expression of T allele relative to C allele ($P < 0.005$) (**Figure 1D**).

CCL2 mRNA transcripts bearing rs13900 C and T alleles have different stability

Previous studies have shown that *CCL2* mRNA is subjected to post-transcriptional regulation through modulation of mRNA stability (Hao and Baltimore, 2009). Here, we determined whether the rs13900 modulates mRNA stability which may in part explain AEI. We used purified monocytes from four heterozygous individuals that were left untreated or stimulated with LPS for 3 h. Cells were either harvested after 3 h of LPS stimulation (considered as t=0) or cultured in the presence or absence of the transcriptional inhibitor actinomycin D for an additional 1, 2, or 4 h. Total RNA was isolated at each time point to assess *CCL2* transcript, calculated as fold induction over unstimulated cells. *CCL2* mRNA levels showed a strong upregulation following LPS stimulation ($P < 0.05$) (**Figure 2A**). Actinomycin D treatment revealed the kinetics of *CCL2* transcript degradation by determining the mRNA half-life, which represents the time (expressed in hours) at which mRNA expression is 50% of the initial level (**Figure 2B**; $t_{1/2}$=Ln (0.5)/slope). For *CCL2*, $t_{1/2}$=1.763 h and is in line with previously published studies that *CCL2* mRNA stability is modulated following inflammatory and cytokine stimuli (Hao and Baltimore, 2009; Zhai et al., 2008).

To rule out the confounding effects of preexisting mRNA, the relative stability of rs13900 C-and T-allele bearing transcripts in heterozygous individuals were evaluated using nascent RNA. Using nascent RNA allows accurate determination of mRNA decay by eliminating the effects of preexisting mRNA. Briefly, monocytes were stimulated with LPS in presence of 5-ethynyl uridine (EU) for 3 hours. Cells were then washed and further incubated with or without Actinomycin D for up to 4 hours. Following Actinomycin D treatment, the nascent RNA was captured after 0, 1, 2 or 4 h using click reaction technique. The click reaction adds a biotin handle to nascent RNA which is then captured by streptavidin beads. cDNA was synthesized from the captured nascent RNA, PCR-amplified and expression of the individual alleles was assessed as described above. While the allelic ratio in the no Act-D samples was ~1.6 in accordance with our initial observation, the allelic ratio was further increased in presence of Act-D. This suggests that post transcriptional mechanisms such as RNA stability may be playing a role in rs13900 mediated CCL2 AEI.

Bioinformatic analyses of rs13900

While previous experimental studies showed that *CCL2* 3' UTR binds HuR, it is not known whether rs13900 disrupts or alters HuR binding (Fan et al., 2011; Lebedeva et al., 2011). Therefore, we analyzed the rs13900 flanking region using various bioinformatic software to mine existing whole genome datasets (e.g., PAR-CLIP datasets) and to predict any mRNA structural changes and altered RBP motifs (**Supplementary Table 2**). We used AURA to examine the colocalization between rs13900 and HuR (**Figure 3A**). As genetic variants can also alter mRNA secondary structure, we used ViennaRNA package to assessed changes in the *CCL2* secondary structure due to rs13900. As shown in **Figure 3B**, rs13900 T allele could potentially alter the *CCL2* transcript secondary structure (red arrow). Further bioinformatic analysis was performed using POSTAR3 suite which incorporates HOMER (Heinz et al., 2010) for motif analysis and RNA context (Kazan et al., 2010) that identifies not only known but also predicts relative binding and structural preferences of RBPs. HOMER motif analysis (Heinz et al., 2010) identified a HuR binding motif in

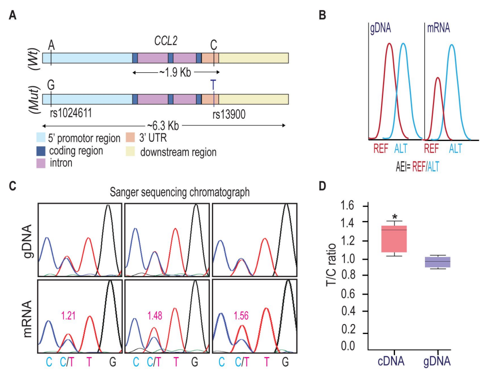


Figure 1.

rs13900 heterozygous individuals exhibit AEI in *CCL2*.

(A) Schematic depicting distal, proximal regulatory elements extending 3 kb on either side of *CCL2* gene and the LD between regulatory polymorphism rs1024611 and the transcribed polymorphism rs13900. rs1024611 is located 2578 base pairs upstream of the *CCL2* translation start site and rs13900 located in the *CCL2* 3' UTR. **(B)** Allelic expression imbalance (AEI) in heterozygous donors is measured as a ratio of alternative allele (ALT) to reference allele (REF) in a transcribed polymorphism. **(C)** Representative chromatograms obtained following Sanger sequencing of PCR products obtained from genomic DNA (gDNA) and reverse transcription-PCR of mRNA (cDNA) from three individuals heterozygous for rs13900. gDNA and mRNA were obtained from PBMC treated with LPS for 3 h as previously described. The allelic ratios shown were determined by PeakPicker analysis. Peakpicker calculates allelic ratios by dividing the peak height of the alternate allele (rs13900 T allele) by that of reference allele (rs13900 C allele). The gDNA peaks were used for normalization. **(D)** Allelic ratio for cDNA and gDNA in six individuals heterozygous for rs13900 after treatment with LPS for 3 h. Statistical significance for the difference in the level of expression between the alleles was determined using Student's *t*-test ($P < 0.003$).

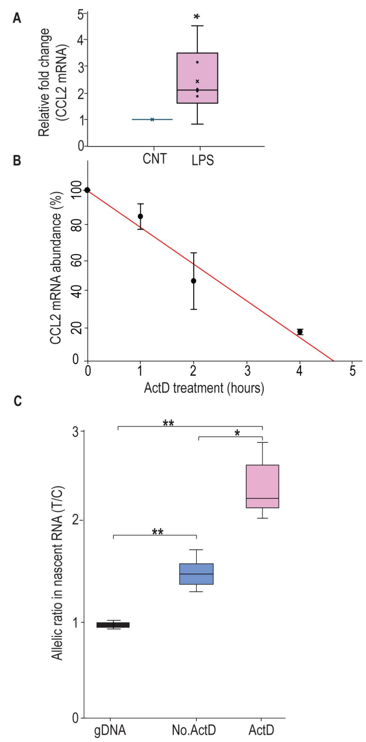


Figure 2.

rs13900T confers greater stability to *CCL2* mRNA.

(A) *CCL2* mRNA expression in peripheral monocytes of heterozygous individuals (n = 6) after treatment with LPS for 3 h and then incubated with 5 µg Act D for indicated times. mRNA was detected by RT-PCR. Results, normalized to 18S rRNA levels, are expressed as folds increase over unstimulated cells (CNT). Levels shown in bar graph represent mean ± SEM of result at time 0 (*P = 0.019) versus unstimulated cells (N = 4). **(B)** *CCL2* mRNA half-life, calculated for each condition as the time (in hours) required for the transcript to decrease to 50% of its initial abundance [t ½ = ln (0.5)/slope]. **(C)** Nascent RNA was isolated from treated monocytes from three individuals in the presence and absence of ActD. Allelic ratio was determined after 4 h of incubation with or without ActD. Expression of rs13900 T allele was much higher in ActD-treated samples. The difference between the groups were assessed by ANOVA with Fisher LSD method (*P < 0.05, ** P < 0.005).

the region flanking the rs13900 (**Fig. 3C**). **Figure 3D** shows the relative structural preference of HuR to different structural contexts identified by RNAcontext, where the letters P, L, U, M indicate that the nucleotide is paired (P), in a hairpin loop (L), in an unstructured (or external) region (U), or Miscellaneous (M). The M category includes various unpaired contexts such as nucleotide localizing to a bulge, internal loop or multiloop. The rs13900 C allele to rs13900 T allele transition is predicted to form a stem (**Figure 3B**) which is predicted to increase HuR binding. The score of 1e was obtained using RBP-Var, a bioinformatics tool that scores variants involved in post-transcriptional interaction and regulation (Mao et al., 2016). Here, the annotation system rates the functional confidence of variants from category 1 to 6. While Category 1 is the most significant category and includes variants that are known to be expression quantitative trait loci (eQTLs), likely affecting RBP binding site, RNA secondary structure and expression, category 6 is assigned to minimal possibility to affect RBP binding. Additionally, subcategories provide further annotation ranging from the most informational variants (a) to the least informational variant (e). Reported 1e denotes that the variant has a motif for RBP binding. Although the employed scoring system is hierarchical from 1a to 1e, with decreasing confidence in the variant's function. However, all the variants in the category 1 are considered potentially functional to some degree. Taken together, our bioinformatic analyses suggested that the rs13900 allele could potentially alter the binding of HuR to *CCL2* 3' UTR.

Differential binding of HuR to rs13900 C and T alleles *in vitro*

To experimentally verify our bioinformatic findings on rs13900, we utilized RNA electrophoretic mobility shift assay (REMSA) to determine whether the region of CCL2 3' UTR that flanks rs13900 binds *in vitro* to HuR and if there are allelic differences in binding. Purified labeled single-stranded oligoribonucleotides corresponding to either rs13900 C or rs13900 T alleles were incubated with 10 µg of whole cell extracts. We tested whether the bound complexes contained HuR by performing antibody mediated supershift assays. As shown in **Figure 4A** and 4B, a predominant shift was observed for the oligoribonucleotide corresponding to rs13900 T allele (lane 8). Notably, there was an approximately 7-fold difference ($P < 0.005$) in oligoribonucleotide/HuR/antibody complexes generated with oligoribonucleotide corresponding to rs13900 T allele which compared to those generated with rs13900 C allele (lane 4). The specificity of the complex formation was confirmed by using a non-specific antibody (**Figure 4A**, lanes 3 & 7). To rule out the possibility that additional RBPs are involved in this complex formation, we used purified HuR protein in mobility shift assays. These studies with purified protein further confirmed that oligoribonucleotides corresponding to rs13900 C or T alleles differentially bound HuR (**Figure 4C**). Taken together, these studies provided experimental validation that the rs13900 may influence differences in binding affinity to HuR (**Figure 4D**).

Differential binding of HuR to rs13900 C and T alleles *ex vivo*

We next tested the hypothesis that the rs13900 is associated with altered binding affinity to HuR *ex vivo*. For this, we performed RNA immunoprecipitation in monocyte/macrophages derived from four heterozygous individuals. Immunoprecipitations were performed using cytoplasmic lysates of macrophages treated with LPS using an affinity purified HuR antibody or IgG. **Figure 5A** shows the relative enrichment of HuR in the immunoprecipitated fraction compared to the IgG control. The HuR immunoprecipitated fraction showed significant enrichment ($P < 0.05$) of the region encompassing rs13900 when compared to the IgG control (**Figure 5B-C**). *CCL2* transcript was enriched ~10-fold in the HuR immunoprecipitated fraction in comparison to the IgG bound fraction (**Figure 5C**). Notably, transcripts corresponding to rs13900 T allele were enriched relative to transcripts containing rs13900 C allele ($P < 0.05$) in the anti-HuR RIP complexes (**Figure 5D**).

Figure 3.

Bioinformatic analysis of the rs13900.

(A) Validation RBP binding sites and polymorphism located on the 3' UTR of *CCL2* transcript from the Atlas of UTR Regulatory Activity and analysis of ENCODE genome-wide data sets detected specific enrichment of HuR (ELAVL1) at the region that contains the rs13900. **(B)** Predicted changes in the secondary structure using Vienna RNA package 2.0; changes in secondary structure are indicated by arrows. **(C)** Sequence logo of HuR binding site as determined by HOMER. **(D)** Relative structural preference of HuR to different structural contexts, the letters P, L, U, M indicate that the nucleotide is paired (P), in a hairpin loop (L), in an unstructured (or external) region (U), and Miscellaneous (M) region.

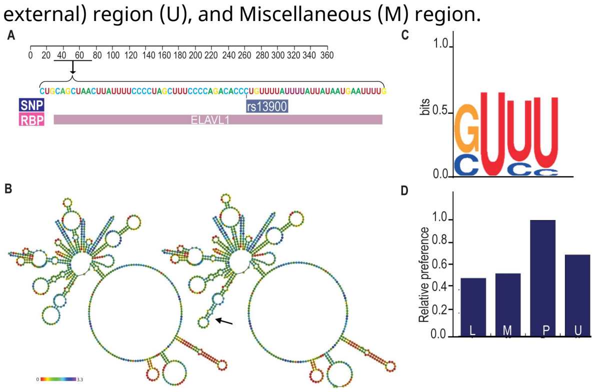
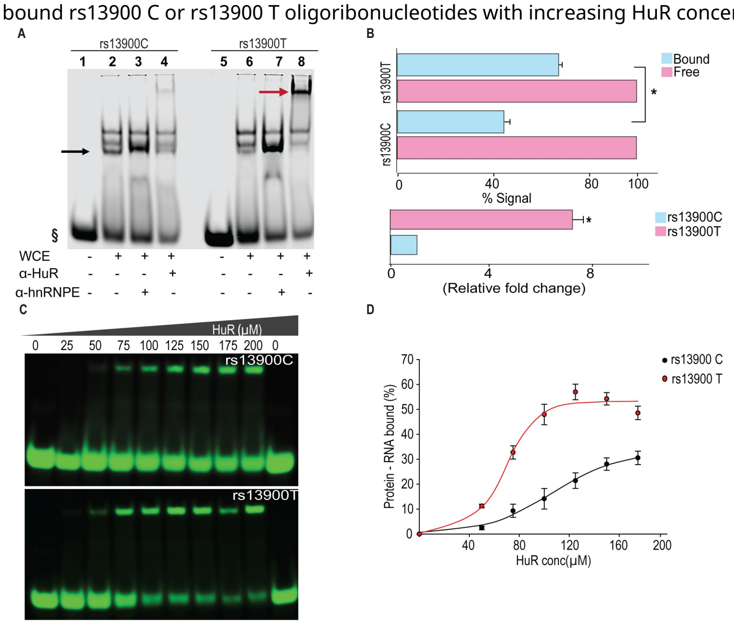


Figure 4.

The rs13900 T allele shows increased in vitro binding of HuR.

(A) REMSA with labelled oligoribonucleotide containing either rs13900 C or T allele and whole cell extracts from 293T cells. § denotes free probe; black arrow, bound probe; red arrow, supershift. **(B)** Representative quantitative densitometric analysis of the antibody shifted complexes suggested increased HuR binding to the oligoribonucleotide bearing rs13900 T allele. The signals in the bound fraction(s) were normalized using the free probe (N = 4). The top panel represents the data from four independent experiments (mean ± SEM). Statistical analyses were performed using Student's t test (**P* < 0.001). The bottom panel shows the relative fold enrichment of the bound protein complexes to the oligoribonucleotide containing the rs13900 T allele relative to that containing the rs13900 C allele . Statistical significance was calculated using Student t test (**P* < 0.001) **(C)** REMSA with labelled oligoribonucleotides containing either rs13900 T or C allele and purified HuR protein. **(D)** Plot showing the fraction of bound rs13900 C or rs13900 T oligoribonucleotides with increasing HuR concentration.



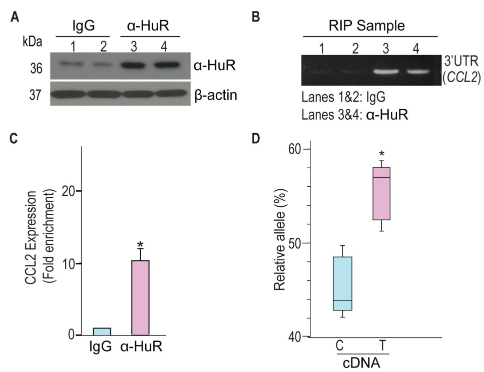


Figure 5.

rs13900 C and T alleles are associated with differential binding to HuR ex vivo.

(A) HuR enrichment in immunoprecipitated material from macrophages stimulated with LPS.
(B) *CCL2* 3' UTR was detected at significant levels in the samples precipitated by α-HuR antibody when compared to the control IgG. **(C)** *CCL2* mRNA expression in anti-HuR antibody enriched immunoprecipitated material analyzed by RT-qPCR (N = 4). Statistical significance was calculated using Student t-test (*$P < 0.005$). The error bars represent SEM. **(D)** Relative expression levels for rs13900 C and T alleles in macrophages stimulated with LPS (N = 4). Statistical significance was calculated using a Student t-test (*$P < 0.005$).

rs13900 T allele confers increased mRNA stability in reporter assays

As HuR is implicated in mRNA stability of many mRNA transcripts including *CCL2*, we tested the hypothesis that *CCL2* 3' UTR influences its stability and that rs13900 modulates this effect. We constructed reporter plasmids that harbor the *CCL2* 3' UTR containing either rs13900 C or rs13900 T allele (**Figure 6A**) and nucleofected them into HEK293 cells as described in Materials and Methods. Luciferase activity was measured after 24 h post-transfection (**Figure 6B**). Our results indicated that the presence of *CCL2* 3' UTR significantly (P < 0.05) reduced the luciferase activity. However, cells transfected with plasmids bearing the rs13900 T allele showed higher luciferase activity when compared with cells transfected with plasmids bearing the rs13900 C allele, suggesting that the presence of rs13900 T allele conferred increased stability to the transcript (P<0.05). We next analyzed the influence of HuR overexpression in reporter vectors containing *CCL2* 3' UTR with either rs13900 C or T alleles. Overexpression of HuR led to a significant increase in luciferase activity of the reporter vector bearing rs13900 T allele (P < 0.05), However, HuR overexpression had no significant effect on the luciferase activity of the reporter vector bearing the rs13900 C allele (**Figure 6C**). Conversely, we examined the effect of HuR on the expression of the reporter assay by co-transfection of HuR siRNA and luciferase reporter constructs. While HuR knockdown had no effect on luciferase activity of the reporter construct bearing rs13900 C allele, it caused a significant reduction in luciferase activity of construct bearing the rs13900 T allele (P < 0.05) (**Figure 6D**). The overexpression and knockdown of the HuR in the nucleofected cells was confirmed using Western blots (**Figure S3**).

Role of HuR-rs13900 interactions in *CCL2* mRNA translatability

The differential interactions with HuR by rs13900 C and rs13900 T alleles could potentially alter *CCL2* mRNA translatability as HuR could increase mRNA translation (Schultz et al., 2020). Therefore, we assessed the relative allelic enrichment in the monosomal and polysomal fractions obtained from MDMs from two individuals exhibiting AEI. Translationally active and inactive pools of RNA were fractionated by isolating the monosomal and polysomal fractions from MDMs on sucrose gradients after the cells were treated with cycloheximide to block translation. The distribution of *CCL2* mRNA in macrophages cultured in presence or absence of LPS for 3 h is depicted in **Figure S4**. As previously reported by others, LPS treatment resulted in a distinct shift of the mRNAs from monosomal fraction to the polysomal fraction (Schott et al., 2014). Consistent with this prior report, 60.4% of the *CCL2* mRNA was associated with polysomal fraction following stimulation of cells with LPS. We assessed the differences in allelic enrichment in cytosolic, monosomal, and polysomic fractions and found that the polysomal fractions showed the enrichment of the rs13900 T allele (**Supplementary Table 3**). However, this differential loading of rs13900 T allele was noted for one donor in cytosolic and monosomal fractions as well.

To further address this question, we used a reporter-based system to assess the effect of rs13900 C and T alleles on the translatability as previously reported (Zhang et al., 2017). To measure translatability, luciferase mRNA and protein were measured simultaneously, and translatability was calculated as luciferase activity normalized by the luciferase mRNA levels after adjusting for protein and 18S rRNA (**Figure 7A-C**). Our results suggest that the rs13900 may alter the mRNA translatability in addition to the transcript stability.

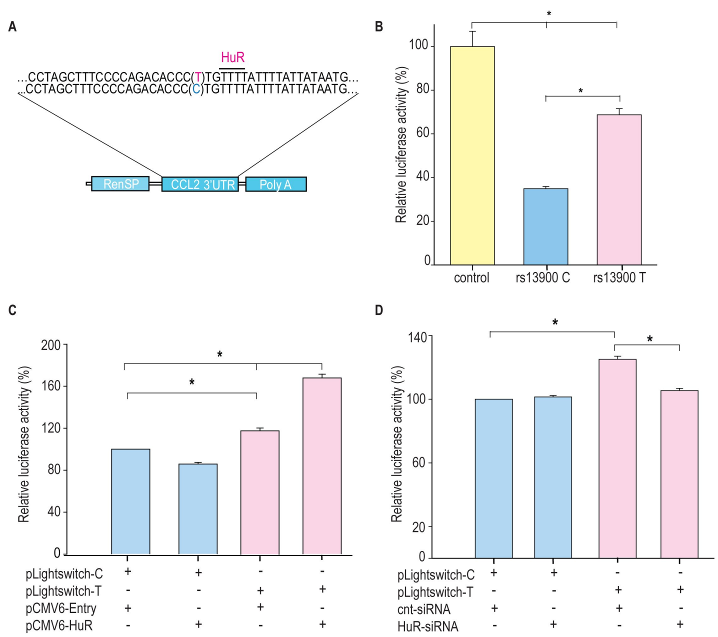


Figure 6.

Differential effects of rs13900 alleles in reporter assays and role of HuR.

(A) Schematic representation of the luciferase reporter vectors containing *CCL2* 3’ UTR with either rs13900 C or T allele. **(B)** HEK-293 cells were transfected with the equal quantities of *CCL2* 3' UTR reporter vectors and luciferase activity was measured 48 hrs later. The relative luciferase activities of the 3' UTR reporter plasmids were expressed as percent reduction in the luminescence when compared to the control vector that was set to 100% after normalizing for the protein content of the lysates. The error bars indicate the standard error of mean and statistical significance was calculated using two-tailed Student’s *t* test (*$P < 0.05$). **(C)** HEK-293T cells were transfected with either pCMV6-HuR (0.5 μg) or pCMV-Entry (0.5 μg) and after 72 hours they were co-transfected with the two plasmid constructs (0.5 μg). Twenty-four hours after transfection the relative change in luciferase activity was determined. **(D)** Cells were co-transfected with 125 pmol HuR siRNA or control siRNA and with the two plasmid constructs (0.5 μg). Twenty-four hours after transfection the relative change in luciferase activity was determined (normalized to total protein concentration, data from four independent experiments (mean ± SEM; N = 4). Statistical analyses were performed using Fisher LSD method (*$P < 0.05$).

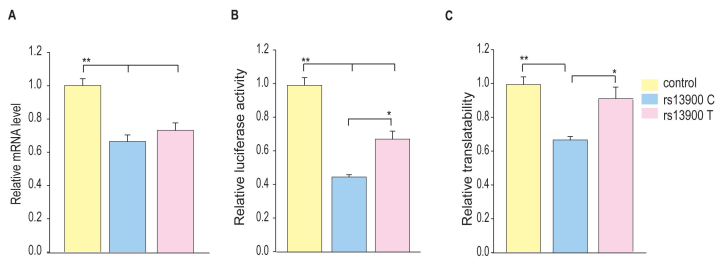


Figure 7.

Influence of rs13900 C and T alleles on *CCL2* translatability.

(A) HEK-293 cells were transfected by nucleofection with control and *CCL2* 3' UTR reporter constructs with rs13900 C and T alleles. The nucleofected cells were plated separately and harvested for total RNA isolation or lysed for mRNA level or protein level expression of luciferase respectively after 24 h. The reporter mRNA levels from the transfected 293 T cells were quantified by qRT-PCR, and 18S rRNA was used for normalization. **(B)** The relative luciferase activities of the 3' UTR reporter plasmids were expressed as a percentage reduction in the luminescence when compared to the control vector that was set to 100% after normalizing for the protein content of the lysates. **(C)** mRNA translatability was calculated as luciferase activity normalized by the reporter luciferase mRNA level. The error bars indicate the standard error of mean from six independent experiments (N = 6) and statistical significance was calculated using ANOVA and post hoc contrast with Fisher LSD method. (**P* < 0.05, ***P* < 0.005).

Differential effect of HuR over-expression on the *CCL2* rs13900 T allele

Our *in vitro* and *ex vivo* data indicate that HuR positively regulates *CCL2* transcript stability by increased binding to the T allele. To determine a direct functional relationship between HuR and *CCL2* AEI, we used a lentiviral overexpression system. We either transduced HuR specific (pCMV6-HuR) or non-specific control shRNAs (CMV-null) into the monocytes obtained from donors who were either homozygous for rs13900 C or T allele. Ready to use GFP-tagged pCMV6-HuR or CMV-null lentiviral particles were transduced into macrophages in the presence of polybrene at an MOI of 1. Cells were processed 72 h following virus addition for the analysis of HuR and *CCL2* mRNA expression. Using this lentiviral system, both high transduction efficiency (90%) and high expression levels were achieved in primary human macrophages (**Figure S5**). Compared with CMV-Null particles, substantial overexpression of HuR was noted with pCMV6-HuR lentiviral particles (**Figure 8A**). HuR overexpression was associated with a higher expression of *CCL2* mRNA in persons homozygous for T allele relative to those who were homozygous for C allele (**Figure 8B**).

Discussion

Genome-wide association studies (GWASs) have led to the discovery of numerous disease susceptibility genetic variants/genes and biological pathways involved in specific diseases, including many with immunological etiologies (Buniello et al., 2019; Manolio, 2010; Visscher et al., 2017). Most of the disease-associated genetic variants identified are present in the non-coding regions of the genome (Maurano et al., 2012; Zhang and Lupski, 2015; Zou et al., 2019). However, characterizing the functional consequence of these genetic variants in the regulatory regions remains a significant challenge to date. About 3.7% of the non-coding variants localize to the untranslated regions (Steri et al., 2018), suggesting post-transcriptional mechanisms such as mRNA stability and translatability could determine disease susceptibility (Maurano et al., 2012).

The *CCL2* locus exemplifies the difficulty in dissecting the functional consequences of cis-regulatory and non-coding genetic variants. Our analysis of regulatory elements in an ~20 kb region upstream of the human *CCL2* coding region identified highly conserved enhancers and other regulatory elements in the *CCL2* locus (Bonello et al., 2011). We identified SNPs with strong LD in this “super-enhancer” and determined their impact on transcriptional activity, which was minimal (Pham et al., 2012). For example, we tested the role of rs7210316 and rs9889296, which had an $r^2 > 0.9$ with rs1024611 in CEU population and were located ~6.3 kb and ~9.2 kb upstream of it (Pham et al., 2012). The genetic regions that encompass and flank these two SNPs either had no or minimal transcriptional activity and lacked the epigenetic markers that have been traditionally associated with gene activation. Additionally, conflicting context-dependent results have been obtained for the transcriptional activities associated with reporter vectors containing rs1024611 A and G alleles, which may have been due to the use of different lengths of the *CCL2* cis-regulatory regions employed in the transcriptional assays. To avoid such context-dependent confounding, we used chromatin annotation, a powerful approach to identify functional SNPs, to generate reporter constructs that span a 6 kb cis-regulatory region (Pham et al., 2012). This approach allowed us to directly compare the roles of four correlated SNPs (rs1860190, rs2857654, rs1024611, and rs2857656) on *CCL2* transcriptional activity. We found no significant differences in transcription strength between these two constructs when transfected into primary human astrocytes (Pham et al., 2012). Furthermore, to understand the basis for increased CCL2 expression associated with the rs1024611 G allele, we and others have demonstrated that several transcription factors bind differentially to *CCL2* polymorphisms,

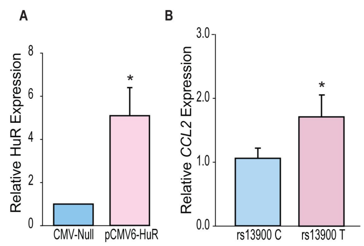


Figure 8.

HuR differentially regulates *CCL2* haplotypes.

(A) HuR expression in primary human macrophages following lentiviral transduction. Macrophages obtained from four individuals who are homozygous for either rs13900 C or rs13900 T allele were transduced with either CMV-null or HuR expressing lentiviral particles (pCMV6-HuR) for 72 hours followed by LPS stimulation for 3 h. **(B)** *CCL2* expression determined by RT-qPCR. Error bars represent SEM. Statistical analyses were performed using Student's t test (* $P < 0.05$).

including IRF-1, PARP-1, STAT-1, Prep1/Pbx complexes (Gonzalez et al., 2002; Mummidi, Bonello and Ahuja, 2009; Page et al., 2011; Wright et al., 2008). However, these previous studies could not unequivocally demonstrate the mechanistic link between differential binding of transcription factors and transcriptional activity and the physiological relevance of implicated transcription factors in CCL2 expression. Our comprehensive studies outlined here demonstrate that rs13900 T allele differentially binds to the RBP HuR, which could alter CCL2 mRNA stability and gene expression. Furthermore, we provide multiple lines of functional evidence for rs13900/HuR role in *CCL2* AEI, including reporter vector assays and HuR overexpression and depletion experiments.

Our previous finding that rs13900 is in perfect LD with rs1024611 has allowed us to exploit the powerful AEI technique and mechanistically link the *CCL2* rs1024611G-rs13900T haplotype to increased expression (Pham et al., 2012). These findings, together with previous knowledge that *CCL2* is subjected to post-transcriptional regulation (Hao and Baltimore, 2009) and regulated by RBP (Fan et al., 2011), raised the possibility that rs13900 could be functional and impact mRNA stability and translatability. In this study, we determined the half-life of *CCL2* to be 1.76 h, reinforcing the fact that post-transcriptional regulation is a key feature of *CCL2* expression (**Fig.2B**). Since transcription and RNA degradation are tightly linked, and transcriptional inhibition may lead to mRNA decay, we assessed the allelic differences in expression in nascent RNA obtained from heterozygous individuals. We found that the *CCL2* transcripts with rs13900 T allele have increased stability relative to those with rs13900 C allele (**Figure 2C**). In addition, both rs1024611 and rs13900 have suggestive associations with *CCL2* expression in *cis*-expression quantitative trait loci (eQTL) analyses. For example, rs1024611 and rs13900 are associated with CCL2 expression in MDM infected with *Listeria monocytogenes* (Nedelec et al., 2016) (−log10(p) = 3.19; Effect size = 0.29±0.084) (Kwong et al., 2021). Remarkably, rs13900 is also identified as a cis-splicing quantitative trait locus (cis-sQTL) for *CCL2* in both untreated and treated monocytes (Alasoo et al., 2018; Kwong et al., 2021; Quach et al., 2016). Recent evidence suggests that genetic variation influencing RNA splicing could play an important role in determining complex phenotypic traits (Li et al., 2016). While rs13900 serves as an sQTL for several *CCL2* transcripts variants, we provide an illustrative example here. rs13900 showed significant associations with the canonical *CCL2* transcript, ENST00000225831 in LPS treated monocytes (−log10(p) = 9.88; Effect size = 0.57±0.084) and with a CCL2 transcript ENST00000580907 (−log10(p) = 9.88; Effect size = −0.57±0.084) that encodes a truncated CCL2 protein (Kwong et al., 2021). While the importance of these associations needs more in-depth studies, our observation that *CCL2* transcripts with rs13900 T allele have a slower degradation could explain the increased CCL2 expression in individuals with rs1024611G-rs13900 haplotype as modest changes in mRNA stability can lead to significant effects on mRNA and protein abundance.

Post-transcriptional gene expression is regulated through interactions between the *cis*-elements in mRNAs and their cognate RBPs (Wu and Brewer, 2012). Studies indicate presence of post-transcriptional operons or regulons – unique subsets of RNAs that associate with RBPs, which coordinate their localization, translation, and degradation. Our bioinformatics analysis and experimental data confirm HuR binds to the region spanning the rs13900. iClip data from HeLa cells (Wang et al., 2010), suggested that the RBPs TIAL1 (T cell intracellular antigen-1 like protein) and U2AF65 (U2 snRNP auxiliary factor large subunit) could also bind to this region of the *CCL2* transcript (Mao et al., 2016). TIAL1 has been proposed to act as a cellular sensor and has been associated with a transcriptome associated with control of inflammation, cell-cell signaling, immune-suppression, angiogenesis, metabolism and cell proliferation (Reyes, Alcalde and Izquierdo, 2009). The role of TIAL1 in *CCL2* expression is not known and additional studies are needed to determine if its binding contributes to *CCL2* AEI. U2AF65 is a widely expressed splicing factor that associates with RNA polymerase II to bind upstream 3' splice sites to facilitate splice site pairing in higher eukaryotes (Hollander et al., 2016). In addition, a change in RNA structure due to a SNP could indirectly alter accessibility of additional regions to RBPs and further studies are required to determine any such changes (Shatoff and Bundschuh, 2020).

The Hu family contains 4 members, of which HuR is ubiquitously distributed. HuR consists of three RNA recognition motifs (RRMs) that are highly conserved and canonical in nature (Ripin et al., 2019). In absence of RNA the three RRMs are flexibly linked but upon RNA binding they transition to a more compact arrangement. Mutational analysis revealed that HuR function is inseparably linked to RRM3 dimerization and RNA binding. Dimerization enables recognition of tandem AREs by dimeric HuR (Pabis et al., 2019) and explains how this versatile protein family can regulate numerous targets found in pre-mRNAs, mature mRNAs, miRNAs and long noncoding RNAs. HuR shuttles between the nucleus and cytoplasm and is likely involved in the transport and stabilization of mRNA. HuR action is antagonized by RNA destabilizing proteins such as TTP (tristetraprolin) and AUF (ARE/poly(U)-binding/degradation factor 1) (Wu and Brewer, 2012). The role of HuR in chemokine gene regulation has been extensively studied in human airway epithelium cells (Fan et al., 2011), they identified that *CCL2* is one of the top targets for HuR and demonstrated that HuR associated with *CCL2* 3' UTR *in vitro* and *CCL2* expression could be modulated by changes in HuR levels using overexpression and RNAi-based experiments. Also, HuR levels influenced *CCL2* mRNA decay. In a mouse model of alcoholic liver disease, HuR was found to play a key role in NOX4 mediated increase in *CCL2* mRNA stability (Sasaki et al., 2017). Notably, HuR has also been implicated in mRNA stability of CX3CL1 (Matsumiya et al., 2010) and IL-8 (Choi et al., 2009) among others. Another study showed that CCL2 itself can induce nuclear to cytoplasmic translocation of HuR and stabilize vascular endothelial growth factor-A (*VEGFA*) mRNA in $CD14^{+}CD16^{low}$ inflammatory monocytes, thus raising the possibility that CCL2 may stabilize its own message in an autocrine fashion (Morrison et al., 2014). These studies further bolster a role for HuR in *CCL2* expression and support our finding that differential binding to HuR alters its expression level.

While there are several examples of genetic variants that influence mRNA stability (Akdeli et al., 2014; Duan et al., 2013; Vilmundarson et al., 2021; Wang, Pitarque and Ingelman-Sundberg, 2006), very few studies have examined the role of RBPs such as HuR in differentially influencing gene expression levels in humans (Steri et al., 2018). A notable exception is the report showing that RBP AUF1 regulates the allele-specific stability of thymidylate synthase (Pullmann et al., 2006). Another study showed that a type 2 diabetes associated polymorphism in the 3' UTR of *PPP1R3* alters distance between two ARE motifs and results in differential binding of protein complexes and may be associated with altered mRNA stability (Xia, Bogardus and Prochazka, 1999). Vilmundarson et al. demonstrated that HuR differentially modulates *IRF2BP2* translation through a 3' UTR polymorphism that is associated with increased coronary artery calcification (Vilmundarson et al., 2021). Our exploratory studies did not resolve whether the rs13900 allelic variation leads to differences in polysomal loading and additional studies are required to address this issue. Another important mechanism by which disease associated genetic variants in the UTRs could influence mRNA stability and translatability is by disrupting or creating microRNA binding sites (Huang et al., 2019). We examined the region flanking the rs13900 and found that there are several predicted binding sites for miRNAs. Among these, several miRNAs have minimum free energy ≤ –25 kcal/mol suggesting that they can potentially play a role in *CCL2* expression and will be investigated in future studies. Of note, a recent study showed that RBPs may play an important role in miRNA mediated gene regulation (Kim et al., 2021).

RBPs regulate the protein expression from a given mRNA by modulation of its half-life, subcellular localization, and ribosomal recruitment (Glisovic et al., 2008). However, mRNA abundance and stability are not always predictive of protein synthesis: relative mRNA abundance/stability and translation levels of a given gene can vary significantly and determined by an assortment of post-transcriptional events (Liu, Beyer and Aebersold, 2016). Our previous findings and results from the present study suggest that there is a close association between *CCL2* rs1024611-rs13900 haplotype, mRNA, and protein expression. Nevertheless, we cannot completely rule out the contribution of transcription-based mechanisms to the allelic expression imbalance at the *CCL2* locus. Previous studies have suggested that there is a high level of conservation for interactions between RBPs and their target molecules and that RNA-mediated gene regulation is less evolvable than transcriptional regulation (Payne, Khalid and Wagner, 2018). Thus, it is remarkable that the rs13900 alters HuR binding and impacts both transcript stability and translatability. Our bioinformatic analysis shows that rs13900 could potentially lead to dramatic changes in the secondary structure of *CCL2* mRNA (**Fig. 3B**). A recent study using bioinformatic analyses has shown that SNPs may affect RNA-protein interactions from outside binding motifs through altered RNA secondary structure (Shatoff and Bundschuh, 2020). We cannot rule out the possibility that HuR may differentially bind to other regions of *CCL2* transcript due to changes in secondary structure and therefore needs to be examined in future studies.

Given our findings that rs13900 modulates HuR binding and thereby influences *CCL2* expression, targeting HuR-ARE interactions could offer a promising therapeutic avenue for conditions involving heightened monocyte/macrophage recruitment, such as inflammatory, cardiovascular, and neoplastic diseases. Indeed, existing small-molecule inhibitors of HuR (e.g., MS-444, KH-3, and CMLD-2) (Chaudhary et al., 2023; Fattahi et al., 2022; Lang et al., 2017; Liu et al., 2020; Wang et al., 2019; Wei et al., 2024) highlight the feasibility of disrupting HuR function in vivo, suggesting that co-targeting HuR and the rs13900 may represent a novel precision-based strategy for monocyte/macrophage-driven disorders.

Both rs1024611 and rs13900 are in high LD with rs3091315, which is detected in the GWA studies of CD and IBD (de Lange et al., 2017; Franke et al., 2010; Kenny et al., 2012; Liu et al., 2015; Palmieri et al., 2010). rs3091315 is classified as an upstream risk variant and is located in the intergenic region between *CCL2* and *CCL7*. The GWA risk allele for CD is rs3091315-A which is correlated with rs1024611(A) and rs13900(C) alleles. While the biological role of *CCL2* in CD and IBD are well recognized (Darkoh et al., 2014; Maharshak et al., 2010; Martin et al., 2019), the results from genetic association studies are not consistent and the disease outcomes could potentially differ by population studied (Chen et al., 2016).

In conclusion, our study shows that the disease associations mediated by the *CCL2* rs1024611-rs13900 haplotype may be due to altered mRNA stability mediated through differential binding of an RBP. Given the importance of mRNA stability in immune homeostasis such mechanisms could play a critical role in inter-individual differences in disease pathogenesis.

Materials and methods

Recruitment of study participants, primary cell culture, and genotyping

All research involving human subjects were approved by the Institutional Review Boards (IRBs) of the University of Texas Health San Antonio, San Antonio, Texas and University of Texas Rio Grande Valley, Edinburg, Texas and Texas A&M University-San Antonio. Written, informed consent from each individual for participation in our study was obtained following the IRB approval. A total of 47 unrelated individuals were recruited into the study. Data and samples from the study participants were obtained at the First Outpatient Research Unit (FORU), UT Health San Antonio, San Antonio, Texas. The rs13900 polymorphism was detected by TaqMan predesigned SNP genotyping assay (Applied Biosystems, CA, USA, Cat. No. 4351379). Briefly, genomic DNA samples were obtained from peripheral blood of healthy study participants using QIAamp DNA Blood Mini Kits (QIAGEN, Cat. No 51104) in accordance with the manufacturer's protocol, checked for quality and concentration, and stored at −80°C until used. Genotyping of rs13900 was performed on 10 ng of genomic DNA using TaqMan genotyping Master mix in a 10 μL reaction volume. PCR was performed on the Quantstudio 12K Flex (Applied Biosystems) in 384-well plates. The temperature cycling conditions consisted of an initial enzyme heat-activation step of 10 min at 95°C and 40 cycles of a 3-step amplification profile of 20 sec at 95°C for denaturation, 1 min at 60°C for annealing, and 30 sec for 72°C for extension. QS12K software (Applied Biosystems, CA, USA). was used to score the alleles. Whole blood samples were collected from study participants who were either homozygous (either C/C or T/T) or heterozygous (C/T) for rs13900 at the initial screening for the genotype status, followed by recalling selected individuals for detailed studies. Peripheral blood mononuclear cells (PBMC) were isolated by a Ficoll density gradient centrifugation. Total genomic DNA and RNA were isolated from $0.5x10^6$ PBMC following stimulation with 1 μg/mL LPS (Millipore Sigma, Cat No. L2630) for 3 h as described previously (Pham et al., 2012; Sharif et al., 2007). CD14+ monocytes were purified using EasySep Human Monocytes Isolation Kit (negative selection kit; Stem cell Technologies, Cat No. 19359) according to manufacturer's instructions. Monocytes were treated with 50 ng/ mL M-CSF (Peprotech, Cat No. 300-25) for 72 h to induce macrophage differentiation, and flow cytometric measurement of surface markers CD64 (BD-Pharmigen, Cat No. 555522), CD206 (BD-Pharmigen, Cat No.564060), and CD44 (BD-Pharmigen, Cat No. 555479) was used to confirm the differentiation (**Figure. S6**).

Real-time quantitative PCR (RT-qPCR) assays

Total RNA was isolated using the RNeasy plus mini kit (Qiagen, Cat No. 74134) according to manufacturer's instructions. 10 μL of total RNA sample (500 ng) was converted to cDNA using MultiScribe Reverse Transcriptase (Applied Biosystems, Cat No. 4311235) with random primers. Real-time quantitative PCR was performed using TaqMan gene expression assays (ThermoFisher Scientific, Cat No. 4331182). Relative expression levels were calculated by applying the $2^{-\Delta\Delta Ct}$ method using 18S rRNA as a reference.

Capture of nascent RNA

Newly synthesized RNA was isolated using the Click-It Nascent RNA Capture Kit (Invitrogen, Cat No: C10365) following the manufacturer's protocol. Peripheral blood mononuclear cells (PBMCs) or monocyte-derived macrophages (MDMs) obtained from heterozygous individuals were stimulated with 1 μg/mL lipopolysaccharide (LPS) for 3 hours, followed by a 3-hour pulse with 0.2 mM 5-ethynyl uridine(Jao and Salic, 2008; Paulsen et al., 2013). After the pulse, the culture medium was replaced with fresh growth medium devoid of EU. To assess RNA stability, cells were left untreated or were treated with actinomycin-D (5 μg/mL) and samples were collected at 0, 1, 2, and 4 h post-treatment. The EU RNA was subjected to a click reaction that adds a biotin handle which was then captured by streptavidin beads. The captured RNA was used for cDNA synthesis (Superscript Vilo kit, Cat No: 11754250), PCR amplification, and allelic quantification.

AEI assessment of *CCL2*

Total RNA and genomic DNA were isolated from primary immune cells (PBMC and macrophages) after they were treated with LPS for 3 h. Total RNA was reverse transcribed as described above. The region encompassing rs13900 was amplified by PCR in a 50 μL reaction mixture containing 2 μL of cDNA or genomic DNA, 25 μL AmpliTaq Gold 360 Master Mix (ThermoFisher Scientific, Cat No: 4398881) and 10 μM each of the forward and reverse primers. The nucleotide sequence of the forward primer and reverse primers were 5'-ACCTGGACAAGCAAACCCAA-3' and 5'-ACCCTCAAAACATCCCAGGG-3'. The following temperature conditions were used: initial denaturation at 95°C for 10 min followed by 40 cycles of denaturation at 95°C for 30 s, annealing at 53.6°C for 30 s, extension at 72°C for 30 s, and final extension at 72°C for 7 min. The amplicons were purified with The GeneJET PCR Purification Kit (ThermoFisher Scientific, Cat No: K0701) and were subjected to Sanger sequencing (Macrogen,USA) using the sequencing primer 5'-GCAAACCCAAACTCCGAAGAC-3'. The degree of AEI for *CCL2* was expressed as a ratio of reference allele (REF; major) to alternative allele (ALT; minor). PeakPicker,v.2.0 was used with default settings to quantify the relative amount of the two alleles from the chromatogram after peak intensity normalization (Ge et al., 2005).

Bioinformatic analyses

The ex vivo binding of RBPs to region flanking rs13900 and *CCL2* 3' UTR was examined using Atlas of UTR Regulatory Activity (AURA) (Dassi et al., 2014). The resources included in POSTAR3 were used to interrogate the RNA binding motifs that overlap the rs13900 (Zhao et al., 2022). RBP-var2 was used to explore the potential functional consequences (Mao et al., 2016), and the ViennaRNA package (Lorenz et al., 2011) was used to predict changes in *CCL2* secondary structure due to rs13900.

RNA electrophoretic mobility shift assay (REMSA)

Oligoribonucleotides spanning the rs13900 C or rs13900 T alleles (IDT) labeled with infrared dyes were used to determine the differential binding of HuR. The sequences of the oligoribonucleotide with rs13900 C allele is rCrUrUrUrCrCrCrCrArGrArCrArCrCr**C**rUrGrUrUrUrUrArUrUrU and oligoribonucleotide with rs13900 T allele is rCrUrUrUrCrCrCrCrArGrArCrArCrCr**U**rUrGrUrUrUrUrArUrUrU. The oligoribonucleotides were incubated with either 10 µg HuR overexpressing HeLa cell whole cell extracts or 25-200 µM ELAVL1 human recombinant protein (Origene, Cat No: TP301562) in a 20 µL reaction mixture with 20 units of RNasin and 1X RNA binding buffer containing 20 mM HEPES (pH 7.6), 3 mM MgCl2, 40 mM KCl, 5% glycerol and 2 mM dithiothreitol, 20 µg yeast tRNA. The reaction was incubated on ice for 10 min and super-shift assay was performed by adding 2 µL of anti-HuR antibody (3A2, mouse monoclonal IgG antibody, Santa Cruz Biotechnology, Cat No: sc-5261). After antibody addition, the complexes were incubated on ice for 15 min and were resolved by electrophoresis under non-denaturing conditions in 5% polyacrylamide gels with 0.5 × TBE running buffer. Gels were then analyzed using Li-Cor Odyssey CLX.

RNA-binding protein immunoprecipitation

RNA-binding protein immunoprecipitation (RIP) was carried out using an immunoprecipitation kit (RIP-assay kit, MBL) following the manufacturer's instructions. Briefly 2x10^6 cells were lysed and the extract was precleared and incubated with 15 µg of antibody against HuR or IgG for 3 h. RNA on the beads was purified and converted to cDNA using random hexamer. cDNA was further amplified and the PCR product sequenced using the following primer: GCAAACCCAAACTCCGAAGAC.

Cell culture and transfection

HEK-293 cell line was maintained in EMEM (ATCC) containing 10% heat-inactivated FBS, penicillin (100 U/mL) and streptomycin (100mg, mL) (GIBCO). Cells were cultured at 37°C in humidified air containing 5% CO2. For HuR overexpression and silencing studies, cells were seeded in six-well plates in serum-free EMEM at a density of ~ 250,000 cells per well. The cells were transfected when they reached ~ 70 to 80% confluency, using lipofectamine 3000 (ThermoFisher Scientific, Cat No: L3000008) according to the manufacturer's recommended protocol. Briefly, lipofectamine 3000 was diluted in OptiMEM and allowed to complex with either 0.5 µg of pCMV6-HuR or pCMV-Entry plasmid or 50 µM of HuR siRNA or control siRNA for 15 min at room temperature. The plasmid/siRNA-lipofectamine 3000 mixture was then added to the cells in a final volume of 2 mL of complete medium. After incubation for 72 h, cells were harvested and used for further processing.

Western blotting

Cell lysates were prepared in chilled RIPA buffer (25 mM Tris-HCL pH 7.6, 150 mM NaCl, 1% NP-40, 1% sodium deoxycholate, 0.1% SDS; Thermo Scientific, Rockford, IL) containing protease inhibitor (complete Mini, EDTA-free Protease inhibitor cocktail tablets, Roche Diagnostics, Cat No: 11836170001). Cells were lysed on ice for 10-15 min followed by centrifugation at 12,000 × g for 15 min, and the cleared supernatant was collected and stored at −20°C. Cell lysates with equal amounts of total protein (15-20 µg) were loaded and separated on a SDS polyacrylamide gel (10–15%) and were electrophoretically transferred onto nitrocellulose membranes (Thermo Scientific). The membranes were blocked with non-fat dry milk in TBST (50 mM Tris-HCl, pH 7.4, 150 mM NaCl, 0.2% Tween 20) and were incubated overnight with anti-HuR primary antibodies (diluted 1:1000). Following incubation, the membranes were washed and then further incubated with anti-mouse ß-actin for one h at room temperature, followed by HRP labeled goat anti-mouse or goat anti-rabbit antibodies (1:1000; Santa Cruz). Protein bands were detected using a chemiluminescence (ECL kit) method (Pierce) and visualized on X-ray film (Kodak).

Reporter assays

Site-specific mutagenesis was done on *CCL2* LightSwitch 3' UTR reporter (Active Motif) to generate constructs containing rs13900 C or rs13900 T alleles. The nucleotide sequence of the complete constructs was verified by Sanger sequencing (Macrogen, USA). 0.5 µg of these constructs were nucleofected into HEK293 cells using 4D Nucleofector (Lonza) and luciferase activity was measured 24 h post transfection using Spectramax plate reader (Molecular Devices). The efficiency of the nucleofection was verified by confocal microscopy, and it was greater than 90%. Given the high efficiency of the nucleofection and to avoid cross-talk with co-transfected control plasmids (Farr and Roman, 1992), luciferase data across samples was normalized using the protein content of the lysates using BCA protein assay. mRNA translatability was assessed using methods as described previously (Zhang et al., 2017).

Polysome profiling

Human monocytes were isolated from fresh blood as described earlier (Gavrilin et al., 2009) with slight modification. Briefly, peripheral blood mononuclear cells were isolated by density gradient centrifugation using Histopaque, followed by immunomagnetic negative selection using EasySep Human Monocyte isolation kit. A high purity level for $CD14^+$ cells was consistently achieved (≥90%) through this procedure, as confirmed by flowcytometry. The purified monocytes were immediately used for macrophage differentiation by treating them with 50 ng/mL M-CSF (PeproTech) for 72 h and flow cytometric measurement of surface markers CD64+, CD206+, CD44 was used to confirm the differentiation (**Fig. S6**). Cells were treated with LPS or PBS for 3 h to induce *CCL2* expression. Cells were treated with 0.1 mM cycloheximide, a translation inhibitor, for 3 min, harvested and washed with ice-cold PBS containing 0.1 mM cycloheximide, and pelleted by centrifugation for 5 min at 500 x g. The pellets were either stored at −80°C or immediately used for cytoplasmic lysate preparation. The cell pellet was resuspended in 500 µL lysis buffer containing 150 mM NaCl, 50 mM Tris-HCl pH 7.5, 10 mM KCl, 10 mM $MgCl_2$, 0.2% NP-40, 2 mM dithiothreitol, 2 mM sodium orthovanadate, 1 mM phenylmethylsulfonyl fluoride, and 80 units/mL RNaseOUT. After 10 min incubation on ice with occasional inverting every two minutes, samples were centrifuged at 12,000 x g for 10 min at 4°C to pellet the nuclei and the post-nuclear supernatants. The optical density at 254 nm was measured, and volumes corresponding to the same OD254 were used. Post-nuclear supernatants were laid on top of a 10-50% sucrose gradient. The gradient was centrifuged for 90 min at 200,000 x g. After the ultracentrifugation, ten fractions were collected from top to bottom. RNA was extracted from each fraction and RT-qPCR was performed using *CCL2* mRNA specific primers. For analysis, fraction one was considered as cytoplasmic lysate, fractions 2-4 were considered monosomal fraction, and fractions 6-10 as polysomal fraction. The percent (%) distribution for the *CCL2* mRNAs across the gradients was calculated using the differences in the cycle threshold (ΔCt) values using the following formula (Nayak et al., 2013): % of mRNA A in each fraction = $2^{\Delta CT \text{ fraction } X}$ x 100/Sum, where ΔCT fraction X = CT (fraction 1) − CT (fraction X)

Lentiviral transduction of primary macrophages

Cells were transfected by slightly modifying the method described by Plaisance-Bonstaff et.al 2019 (Plaisance-Bonstaff et al., 2019). Briefly, monocytes were purified from PBMCs obtained from homozygous donors for rs13900 C or rs13900T by negative selection. Upon purification cells were resuspended in 24 well plates at a seeding density of 0.5 x10^6 cells per well and were further cultured in the medium supplemented with 50 ng/mL M-CSF (**Figure. S7**). After 24 h, ready to use GFP-tagged pCMV6-HuR or CMV-null lentiviral particles (Amsbio, Cambridge, M.A) were transduced into 0.5 x10^6 cells in presence of polybrene (60 µg/mL) at a MOI of 1. The cells were processed for HuR and *CCL2* expression 72 h after transduction after stimulation with LPS for 3 h.

Statistical Analyses

All values in figures are presented as the mean ± SEM. Results were analyzed with ANOVA and posthoc contrasts with Fisher (Least Significant Difference (LSD) test or Student's t test. Statistical analyses were carried out in the Sigma Plot 12.0 software.

Web resources

AURA, *http://aura.science.unitn.it/* POSTAR3, *http://postar.ncrnalab.org*

RBP-var2, *http://159.226.67.237/sun/RBP-Var/index.php/Rbp/index* ViennaRNA package, *https://www.tbi.univie.ac.at/RNA/*

Supplementary figures and tables

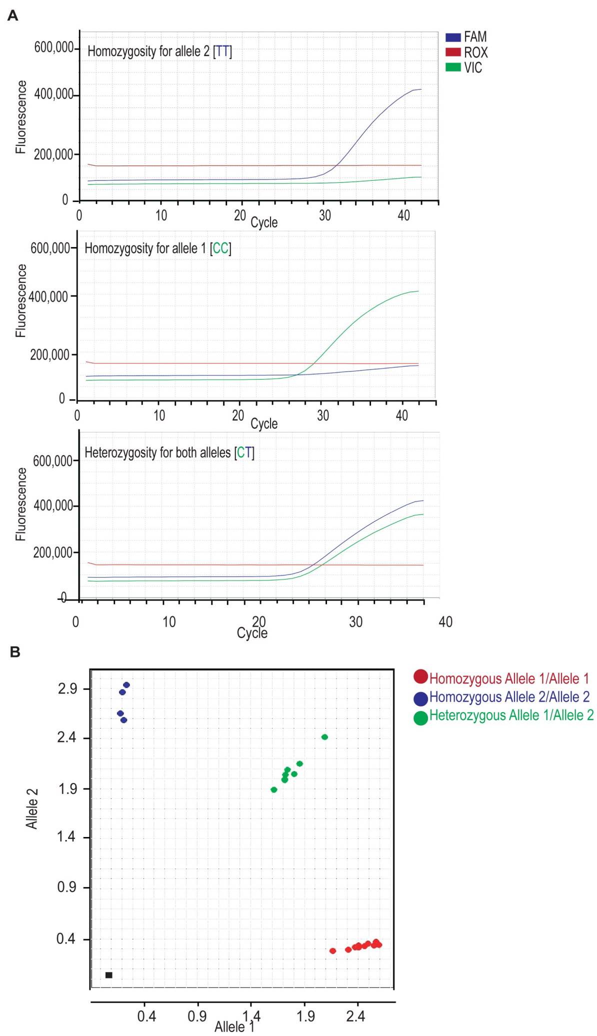


Figure S1.

SNP genotyping for rs13900 using TaqMan technology.

(A) Real-time multicomponent amplification curves for the TaqMan's rs13900 assay. Fluorescence signal detected for individuals with rs13900T, rs13900C and rs13900CT genotypes. The blue curve indicates amplification of T allele (alternate/mutant allele) while the green curve represents C allele (wild type). **(B)** Allelic discrimination plot generated on Quant Studio 12K Flex Real-Time PCR system showing higher signal and better cluster separation for the rs13900 assay. Allele C is plotted on X-axis and the T allele is plotted on Y-axis.

Figure S2.

Time course of *CCL2* mRNA expression in PBMCs.

Peripheral blood mononuclear cells (PBMC) were treated with 1 μg lipopolysaccharide (LPS) for 1, 3 or 6 hours. *CCL2* mRNA expression was measured by quantitative real-time PCR and normalized to 18S rRNA. Data are presented as mean ± SD of triplicate samples.

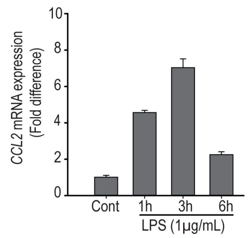


Figure S3.

Validation of HuR overexpression and silencing by Western blotting.

(A) A representative western blot showing HuR protein levels in HEK293 cells transfected with either pCMV6-HuR plasmid or HuR-targeting siRNA (HuR-siRNA). **(B)** Densitometric analysis of Western blot bands. The histogram shows relative intensity of HuR bands in each sample, normalized to β-actin. Error bar represents standard error mean from three independent experiments. *p<0.025, ** p<0.05.

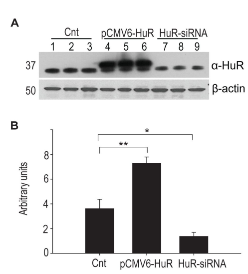


Figure S4.

Polysomal association of *CCL2* mRNA before and after LPS stimulation of macrophages

(A) Polysome profile obtained by sucrose gradient centrifugation from macrophages before and after stimulation with LPS (1 μg/mL) for 3 h. The polysome profile shows a shift from monosome to heavier polysome upon LPS stimulation, indicating active translation. **(B)** The percentage of *CCL2* mRNA loading on polysome fraction was calculated using ΔCT method (mean ± SEM, n=4). *p<0.025.

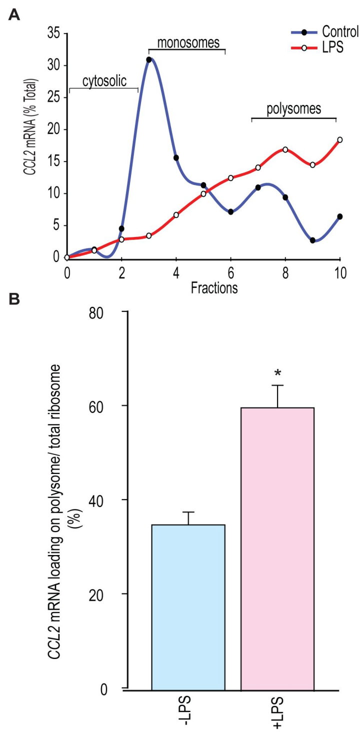
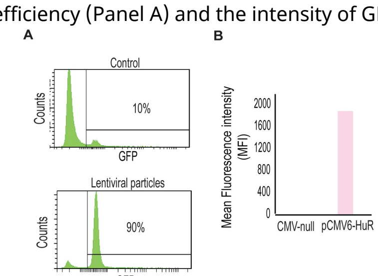


Figure S5.

Transduction efficiency of pCMV6-HuR.

Flow-cytometric analysis of transduction efficiency (Panel A) and the intensity of GFP in transduced cells (Panel B).



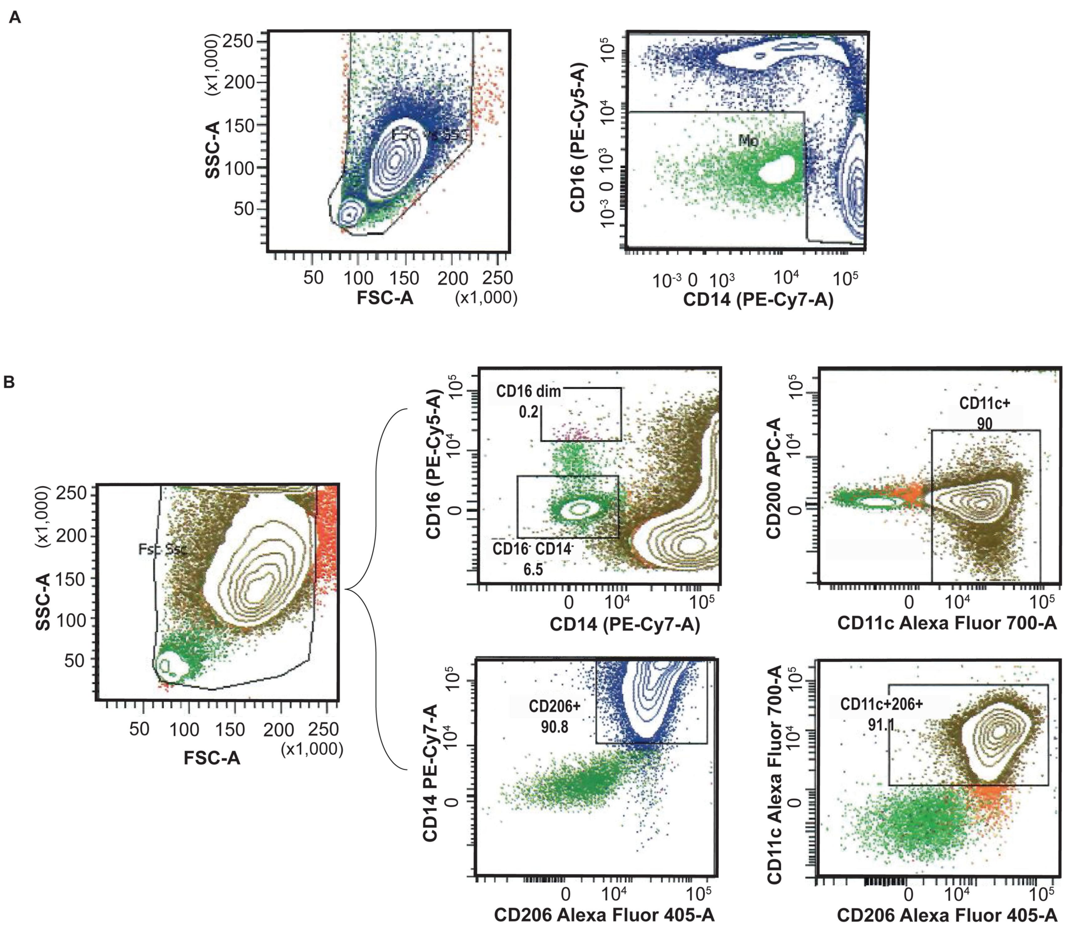


Figure S6.

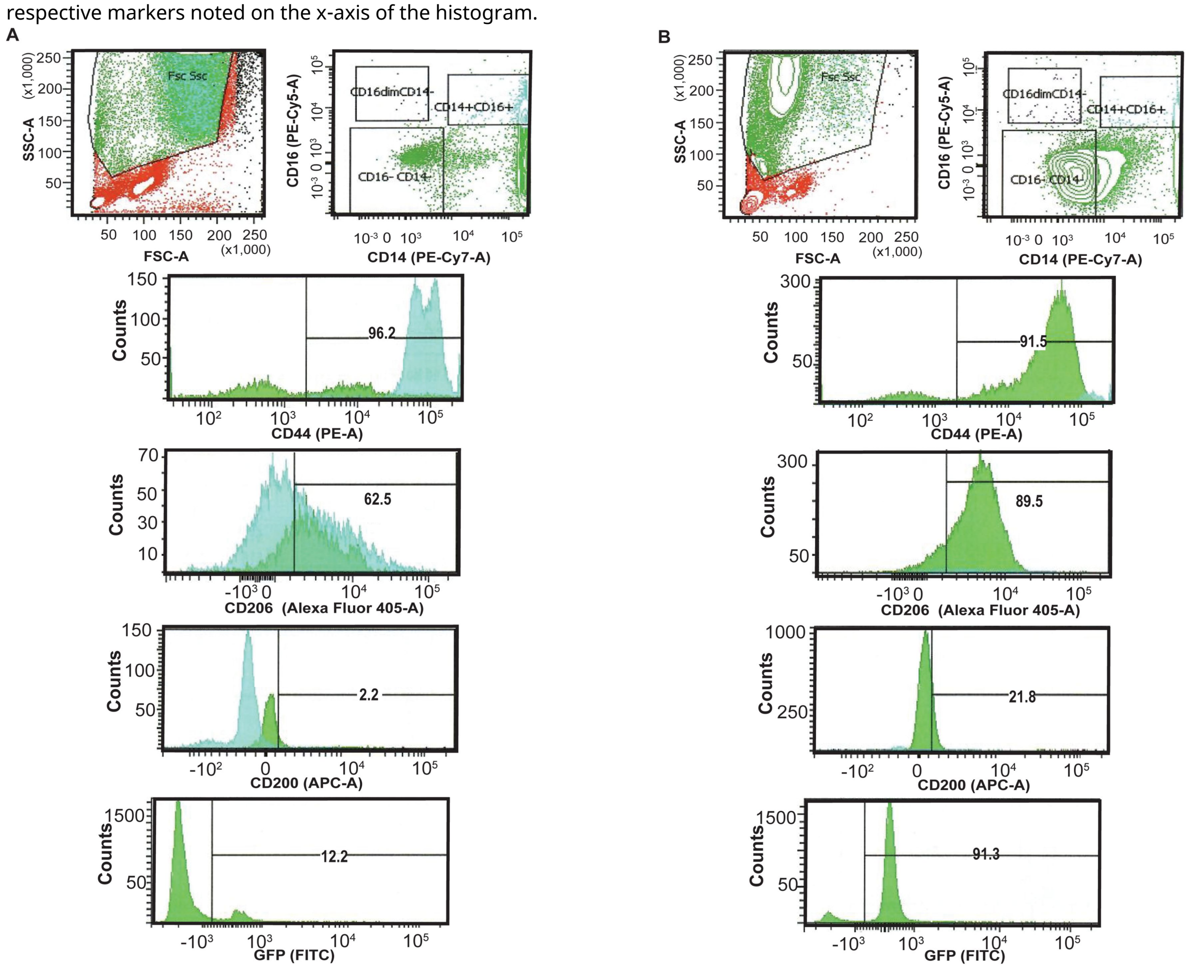
Flow cytometry analysis of cell surface markers to assess the purity of monocytes isolated from PBMC and in vitro differentiation to macrophages.

(A) A representative image depicting the purity of monocytes isolated from PBMCs by immunomagnetic negative selection. A high monocyte content (>90%) was achieved in the enriched fraction. **(B)** Multicolor flow cytometry to determine macrophage differentiation. The macrophage population was identified by forward and side scatter and immunostaining. Surface expressions of CD11c, CD206, CD200, CD16 and CD14 were measured.

Figure S7.

Lentiviral transduction of differentiated macrophages.

The differentiation of monocytes into macrophages and their transduction was confirmed by detection of GFP and the cell surface markers CD44, CD200, CD206 by flow cytometry. Cells were transduced with either GFP-tagged pCMV-null **(A)** or GFP-tagged pCMV-HuR particles at MOI of 1 **(B)**. Each bracketed region in the histogram corresponds to the % of positive cells for respective markers noted on the x-axis of the histogram.



Supplementary Table 1.

Allele carriages and allele frequencies of rs13900 in healthy volunteers, MAF, minor allele frequency.

	rs13900 (n=42)
Reference allele [CC]	18 (42.8%)
Heterozygous [CT]	16 (38.09%)
Alternative allele [TT]	8 (19.04%)
MAF	0.38
Heterozygosity	0.47
No of alleles	84
Hardy Weinberg *P*	0.459

Supplementary Table 2.

Crosslinking immunoprecipitation (CLIP) analysis of HuR binding sites on the 3’untranslated region (3’UTR) of the *CCL2* gene.

Summary of the SNPs located within HuR-binding regions is shown including SNP ID (rs#), genomic coordinates, strand, binding score, conservation scores (PhastCons, PhyloP), dataset accession numbers, alleles, and SNP position within peak region. Data was generated using POSTAR3 using the variation module.

RBP	Tissue type	rs#	Position	Strand	CLIP-seq technology and peak calling method	Score	PhastCons score	PhyloP score	Data accession	Ref strand	Alt strand	SNP position
ELAVL1	HeLa	rs13900	chr17:34256881-34256901	+	PAR-CLIP,Piranha_0.01	11	0.024	0.227	GSE29943,GSM741175	C	T	chr17:34256891-34256892
ELAVL1	HeLa	rs181021073	chr17:34256857-34256876	+	PAR-CLIP,PARalyzer	0.647	0.002	0.14	GSE29943,GSM741173	T	G	chr17:34256868-34256869
ELAVL1	HeLa	rs181021073	chr17:34256857-34256876	+	PAR-CLIP,PARalyzer	0.653	0.002	0.14	GSE29943,GSM741174	T	G	chr17:34256868-34256869
ELAVL1	HeLa	NA	chr17:34257108-34257130	+	PAR-CLIP,PARalyzer	0.623	0.001	-0.256	GSE29943,GSM741173	AT	A	chr17:34257109-34257110
ELAVL1	HeLa	NA	chr17:34257101-34257121	+	PAR-CLIP,Piranha_0.01	9	0	-0.39	GSE29943,GSM741173	AT	A	chr17:34257109-34257110
ELAVL1	HeLa	NA	chr17:34257101-34257121	+	PAR-CLIP,Piranha_0.01	6	0	-0.39	GSE29943,GSM741175	AT	A	chr17:34257109-34257110

Supplementary Table 3.

Loading of rs13900 alleles to cytosolic, monosomal and polysomal fractions from macrophage extracts prepared from heterozygous donors.

A T:C ratio > 1 indicates increased levels of the T allele relative to C allele.

	gDNA	Cytosol	Monosome	Polysome
D-1	0.975806	0.872946	0.873425	1.778429
D-2	1.034091	1.557778	1.441919	1.562622

S.N	Reagents	Source	Cat.No.	Note
1	rs13900 TaqMan Assay	ThermoFisher Scientific	4351379	C7449810_10*
2	QIAamp DNA Blood Mini Kit	Qiagen	51104	
3	TaqMan Genotyping Master Mix	ThermoFisher Scientific	4371355	
4	Histopaque-1077	Millipore Sigma	10771	
5	Lipopolysaccharides from *E.coli* 0:111:B4	Millipore Sigma	L2630	1μg/mL**
6	EasySep Human Monocyte Isolation Kit	StemCell Technologies	19359	
7	Recombinant Human M-CSF	Peprotech	300-25	50 ng/mL**
8	Pe-Cy 7 Mouse-Anti Human CD14$^+$	BD-Pharmigen	562698	
9	FITC Mouse- Anti Human CD64+	BD-Pharmigen	555522	
10	BV421 Mouse Anti-Human CD206+	BD-Pharmigen	564060	
11	AlexaFluor 700 Mouse-Anti Human CD11c$^+$	BD-Pharmigen	561352	
12	PE CY 5 Mouse Anti-Human CD16+	BD-Pharmigen	555408	
13	RNeasy Plus Mini kit	Qiagen	74134	
14	MultiScribe Reverse Transcriptase	ThermoFisher Scientific	4311235	
15	CCL2 TaqMan Gene Expression Assays	ThermoFisher Scientific	4331182	Hs00736046_m1*
16	18S TaqMan Gene Expression Assays	ThermoFisher Scientific	4331182	Hs03003631_g1*
17	Click-iT-Nascent RNA Kit	Invitrogen	C10365	
18	Actinomycin D	MilliporeSigma	A1410	
19	Superscript Vilo Kit	ThermoFisher Scientific	11754250	
20	AmpliTaq Gold 360 Master Mix	ThermoFisher Scientific	4398881	
21	GeneJET PCR Purification Kit	ThermoFisher Scientific	K0701	
22	ELAVL1 human recombinant protein	OriGene	TP301562	
23	HuR/ELAV1 Antibody	Santa Cruz	sc-5261	
24	hnRNP E1 Antibody (T-18)	Santa Cruz	sc-16504	
25	Normal Goat IgG	Santa Cruz	sc-2028	
26	β Actin Antibody (2A3)	Santa Cruz	sc-517582	
27	Goat Anti-Mouse IgG-HPR	Santa Cruz	sc-2005	
28	Lipofectamine 3000	ThermoFisher Scientific	L3000008	
29	RIPA Lysis and Extraction Buffer	ThermoFisher Scientific	89900	
30	Protease Inhibitor Cocktail Tablets	Millipore Sigma	11836170001	
31	Nitrocellulose Membranes	Thermo Scientific	88024	
32	RIP-Assay Kit	MBL Life science	RN1001	

Supplementary Table 4.

Reagents used in the study with source and catalog numbers.

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Additional information

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Author information

Feroz Akhtar
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

Joselin Hernandez Ruiz
Utah Center for Genetic Discovery, Department of Human Genetics, University of Utah, Salt Lake City, United States

Ya-Guang Liu
Department of Pathology, School of Medicine, University of Texas Health San Antonio, San Antonio, United States

Roy G Resendez
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

Denis Feliers[‡]
Department of Medicine, School of Medicine, University of Texas Health San Antonio, San Antonio, United States

[‡]Present address: Novartis Institute for Biomedical Research San Diego, San Diego, United States.

Liza D Morales
South Texas Diabetes and Obesity Institute, Department of Genetics, School of Medicine, University of Texas Rio Grande Valley, Brownsville, United States

Alvaro Diaz-Badillo
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

Donna M Lehman
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

Rector Arya
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

Juan Carlos Lopez-Alvarenga
Department of Population Health and Biostatistics, School of Medicine, University of Texas Rio Grande Valley, Harlingen, United States
ORCID iD: 0000-0002-0966-8766

John Blangero
South Texas Diabetes and Obesity Institute, Department of Genetics, School of Medicine, University of Texas Rio Grande Valley, Brownsville, United States
ORCID iD: 0000-0001-6250-5723

Ravindranath Duggirala
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

Srinivas Mummidi
Department of Health and Behavioral Sciences, Texas A&M University-San Antonio, San Antonio, United States

For correspondence: srinivas.mummidi@tamusa.edu

Editors

Reviewing Editor
Detlef Weigel
Max Planck Institute for Biology Tübingen, Tübingen, Germany

Senior Editor
Detlef Weigel
Max Planck Institute for Biology Tübingen, Tübingen, Germany

Reviewer #1 (Public review):

Summary:

This paper presents evidence that a relatively common genetic variant tied to several disease phenotypes affects the interaction between the mRNA of CCL2 and the RNA binding protein HuR. CCL2 is an immune cell chemoattractant protein.

Strengths:

The study is well conducted with relevant controls. The techniques are appropriate, and several approaches provided concordant results were generally supportive of the conclusions reached. The impact of this work, identifying a genetic variant that works by altering the binding of an RNA-regulatory protein, has important implications given that the HuR protein could be a drug target to improve its function and over-ride this genetic change. This could have important implications for a number of diseases where this genetic variant contributes to disease risk.

The authors have done a nice job of citing prior work. Details of the experimental protocols are well elaborated and the significance of the findings are well contextualized.

Weaknesses:

Authors have addressed prior weaknesses.

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Reviewer #2 (Public review):

This study focuses on the differential binding of the RNA-binding protein HuR to CCL2 transcript (genetic variants rs13900 T or C). The study explores how this interaction influences the stability and translation of CCL2 mRNA. Employing a combination of bioinformatics, reporter assays, binding assays, and modulation of HuR expression, the study proposes that the rs13900T allele confers increased binding to HuR, leading to greater mRNA stability and higher translational efficiency. These findings indicate that rs13900T allele might contribute to heightened disease susceptibility due to enhanced CCL2 expression mediated by HuR. The study is interesting and most results are convincing, however the interpretation relative to RNA transcription and/or stability must be modified, and some data need better presentation or interpretation.

Major Points

Figure 2C:
The authors describe an experiment to assess mRNA stability by labeling nascent RNA with EU for 3 hours, followed by washout of EU, and then incubation with or without actinomycin

D for an additional 4 hours before measuring the remaining EU-labeled RNA. While the approach to label nascent RNA with EU is appropriate for tracking RNA decay, I have concerns regarding the use and interpretation of actinomycin D in this context.
After EU washout, the pool of EU-labeled RNA is fixed and no new EU incorporation can occur. Therefore, the addition of actinomycin D at this stage should not affect the decay rate of the already labeled RNA, as transcription of EU-labeled RNA has effectively ceased. In this design, measuring the decrease in EU-labeled RNA over time reflects mRNA stability (even in absence of actinomycin D) rather than transcriptional activity.
Therefore, the authors' statement that the non-actinomycin D treatment group represents transcriptional changes is not accurate here. Since EU labeling was stopped prior to the 4-hour incubation, any changes in EU-labeled RNA levels during this period reflect RNA decay, not new transcription.

In summary:
To assess transcriptional changes, one would compare the amount of EU-labeled RNA synthesized during the initial labeling period (the first 3 hours), before washout.
If the authors wish to use actinomycin D to block transcription, this should be done in a separate decay assay without EU labeling.
In the current experimental setup, actinomycin D is unnecessary after EU washout and does not influence the decay of the labeled RNA.
I recommend the authors reconsider the interpretation of their data accordingly. I recommend to remove the data points relative to the presence of actinomycin D, as the non-actinomycin D samples are already representative of post-transcriptional changes given that EU was washed out. If Authors want to assess transcriptional changes, they would have to assess the levels during the initial labeling period (before the washout). Transcriptional differences were not assessed, therefore I would modify the text accordingly.
In this context, any changes observed in the actinomycin D-treated samples are likely attributable to general cellular stress induced by actinomycin D, which is known to be highly stressful for cells. This stress could indirectly influence the decay rates of already-labeled EU-RNA.

Figure 4C and 4D:
The Author provided an updated gel with relative quantification - which effectively show the enhanced binding of CCL2 mRNA carrying the T variant to HuR - but they only provided it as data for reviewers (Figure R1). I highly recommend to use these data in the final manuscript instead of the data currently presented in Figure 4C and 4D. This would be important in order not to not create confusion in the reader or concerns regarding probe degradation or saturation.

Minor points
For the IP, I recommend to explain in the final version why the input was not provided (lack of material) and to clarify that the specific binding of Actin was used as a loading control in absence of input. This would be highly beneficial for the readers.

https://doi.org/10.7554/eLife.93108.2.sa1

Author response:

The following is the authors’ response to the original reviews

> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Comment 1: The authors need to do more to cite the prior work of others. CCL2 allelic expression imbalance tied to the rs13900 alleles was first reported by Johnson et al. (Pharmacogenet Genomics. 2008 Sep; 18(9): 781-791) and should be cited in the*

> *Introduction on line 128 next to the Pham 2012 reference. Also, in the Results section, line 142, please provide references for the statement "We and others have previously reported a perfect linkage disequilibrium between rs1024611 in the CCL2 cis-regulatory region and rs13900 in its 3' UTR" since the linkage disequilibrium for these 2 SNPs is not reported in the ENSEMBL server for the 1000 genomes dataset. #*

We thank the reviewer for pointing out the omission regarding the citation of prior work. We acknowledge that Johnson et al. (2008) reported the association between rs13900 and *CCL2* allelic expression imbalance based on Snapshot methodology while examining _cis-_acting variants of 42 candidate genes. To acknowledge these prior studies, we have cited the previous works of Johnson et al. (Johnson et al., 2008) along with Pham et al. (Pham et al., 2012) that linked rs13900 to CCL2 allelic expression imbalance. The text in the introduction section (Lines 128-130) has been updated to reflect the above-mentioned changes.

“We and others have demonstrated AEI in CCL2 using rs13900 as a marker with the T allele showing a higher expression level relative to C allele (Johnson et al., 2008; Pham et al., 2012).”

We have cited some previous studies that suggested strong linkage disequilibrium between rs1024611 and rs13900 within *CCL2* gene, with D’=1 and R^2=0.96 (Hubal et al., 2010; Intemann et al., 2011; Kasztelewicz et al., 2017; Pham et al., 2012) on Line 144. To address the concern regarding unreported linkage disequilibrium between rs1024611 and rs13900, we reviewed the pairwise linkage disequilibrium data by population in the ENSEMBL server for 1000 Genome dataset and confirm that the linkage disequilibrium (LD) between rs1024611 and rs13900 has been observed, with D’=1 and R^2=0.92 to 1.0 in specific populations. We have included a table (Author response table 1) depicting pairwise LD between rs13900 and rs1024611 as reported in the ENSEMBL server for the 1000 genome dataset, a URL reference to the ENSEMBL server data.

Author response table 1.

Pairwise linkage disequilibrium data between rs13900 and rs1024611 by population reported in the ENSEMBL server for the 1000 genome dataset

Population	Description	Focus Variant	Variant 2	r2	D'
1000GENOMES:phase_3:PEL	Peruvian in Lima, Peru	rs1024611	rs13900	0.92478	1
1000GENOMES:phase_3:YRI	Yoruba in Ibadan, Nigeria	rs1024611	rs13900	0.97254	1
1000GENOMES:phase_3:GWD	Gambian in Western Division, The... (more)	rs1024611	rs13900	0.9737	1
1000GENOMES:phase_3:FIN	Finnish in Finland	rs1024611	rs13900	0.97835	1
1000GENOMES:phase_3:JPT	Japanese in Tokyo, Japan	rs1024611	rs13900	0.97905	1
1000GENOMES:phase_3:CHB	Han Chinese in Bejing, China	rs1024611	rs13900	0.97919	1
1000GENOMES:phase_3:ACB	African Caribbean in Barbados	rs1024611	rs13900	1	1
1000GENOMES:phase_3:ASW	African Ancestry in Southwest US	rs1024611	rs13900	1	1
1000GENOMES:phase_3:BEB	Bengali in Bangladesh	rs1024611	rs13900	1	1
1000GENOMES:phase_3:CDX	Chinese Dai in Xishuangbanna, China	rs1024611	rs13900	1	1
1000GENOMES:phase_3:CEU	Utah residents with Northern and... (more)	rs1024611	rs13900	1	1
1000GENOMES:phase_3:CHS	Southern Han Chinese, China	rs1024611	rs13900	1	1
1000GENOMES:phase_3:CLM	Colombian in Medellin, Colombia	rs1024611	rs13900	1	1
1000GENOMES:phase_3:ESN	Esan in Nigeria	rs1024611	rs13900	1	1
1000GENOMES:phase_3:GBR	British in England and Scotland	rs1024611	rs13900	1	1
1000GENOMES:phase_3:GIH	Gujarati Indian in Houston, TX	rs1024611	rs13900	1	1
1000GENOMES:phase_3:IBS	Iberian populations in Spain	rs1024611	rs13900	1	1
1000GENOMES:phase_3:ITU	Indian Telugu in the UK	rs1024611	rs13900	1	1
1000GENOMES:phase_3:KHV	Kinh in Ho Chi Minh City, Vietnam	rs1024611	rs13900	1	1
1000GENOMES:phase_3:LWK	Luhya in Webuye, Kenya	rs1024611	rs13900	1	1
1000GENOMES:phase_3:MSL	Mende in Sierra Leone	rs1024611	rs13900	1	1
1000GENOMES:phase_3:MXL	Mexican Ancestry in Los Angeles... (more)	rs1024611	rs13900	1	1
1000GENOMES:phase_3:PJL	Punjabi in Lahore, Pakistan	rs1024611	rs13900	1	1
1000GENOMES:phase_3:PUR	Puerto Rican in Puerto Rico	rs1024611	rs13900	1	1
1000GENOMES:phase_3:STU	Sri Lankan Tamil in the UK	rs1024611	rs13900	1	1

F. Variant, Focus Variant; R^2, correlation between the pair loci; D’, difference between the observed and expected frequency of a given haplotype.

URL: https://www.ensembl.org/Homo_sapiens/Variation/HighLD?db=core;r=17:34252269-34253269;v=rs1024611;vdb=variation;vf=959559590;second_variant_name=rs13900

> *Comment 2: Certain details of the experimental protocols need to be further elaborated or clarified to contextualize the significance of the findings. For example, in the results line 184 the authors state "Using nascent RNA allows accurate determination of mRNA decay by eliminating the effects of preexisting mRNA." How does measuring nascent RNA enable the accurate determination of mRNA decay? Doesn't it measure allele-specific mRNA synthesis? Please elaborate, as this is a key result of the study. Can the authors provide a reference supporting this statement?*

It is worthwhile to mention that mRNA decay can be precisely measured by eliminating the effect of any preexisting mRNA. Metabolic labeling with 4-thiouridine allows exclusive capture of newly synthesized RNA which will allow quantification of RNA decay eliminating any interference from preexisting RNA. We agree that nascent RNA measurement primarily reflects synthesis rate rather than degradation. However, in conjugation with actinomycin-D based inhibition studies it can be exploited for accurate mRNA decay determination of the newly synthesized RNA (Russo et al., 2017). Therefore, our aim was to use the nascent RNA to study decay kinetics. The imbalance in the *CCL2* allele expression does occur at the transcriptional level as seen in non-actinomycin-D treatment group (Figure 2C) although the impact of post-transcriptional mechanisms that alter transcripts stability cannot be ruled out. Therefore, we employed a novel approach that could assess both the synthesis and the degradation by combining actinomycin-D inhibition and nascent RNA capture in the same experimental setup. In the presence of actinomycin-D, we could detect much greater allelic difference in the expression levels of the rs13900T and C allele four-hour post-treatment, suggesting a role for post-transcriptional mechanisms in CCL2 AEI.

“We have expanded the method section in the revised draft to include experimental details on capture of nascent RNA and subsequent downstream analysis” (Lines 553-563).

Newly synthesized RNA was isolated using the Click-It Nascent RNA Capture Kit (Invitrogen, Cat No: C10365) following the manufacturer’s protocol. Peripheral blood mononuclear cells (PBMCs) or monocyte-derived macrophages (MDMs) obtained from heterozygous individuals were stimulated with lipopolysaccharide (LPS) for 3 hours in presence of 0.2 mM 5-ethynyl uridine (EU) (Jao and Salic, 2008; Paulsen et al., 2013). After the pulse, the culture medium was replaced with fresh growth medium devoid of EU. To assess RNA stability, actinomycin-D (5 µg/mL) was added, and samples were collected at 0, 1, 2, and 4 h post-treatment. The EU RNA was subjected to a click reaction that adds a biotin handle which was then captured by streptavidin beads. The captured RNA was used for cDNA synthesis (Superscript Vilo kit, Cat No: 11754250), PCR amplification, and allelic quantification.”

> *Comment 3: Also, they next state that the assay was carried out using cells treated with actinomycin D (line 186). Doesn't actinomycin D block transcription? The original study by Jia et al 2008 in PNAS reported that low concentration of ActD (100 nM) blocked RNA pol I and higher concentration (2 uM) blocked RNA pol II. This or the study on which the InVitrogen kit is based should be cited. The concentration of actinomycin D used to treat the cells should be given. They report that the T allele transcript was more abundant than the C allele transcript in nascent RNA. Why doesn't that argue for a transcriptional mechanism rather than an RNA-stability mechanism? This result should be discussed in the Discussion.*

In our study, we used a concentration of 5 µg/mL (3.98 µM), which as noted by the reviewer can effectively inhibit RNA polymerase II (Pl II) activity. We have updated our manuscript to include details and cited the original work of (Jao and Salic, 2008; Paulsen et al., 2013), which thoroughly investigate the effect of various concentrations of ActD on RNA polymerase I and II (Line no 557). A discussion of the RNA stability mechanism is provided in the Result section (Lines 196-198).

> *Comment 4: In their bioinformatics analysis of the allele-specific CCL2 mRNAs, they reported that the analysis obtained a score of 1e (line 214). What does that mean? Is it significant?*

We acknowledge that the notation “a score of 1e” was unclear and thank the reviewer for pointing it out. We have clarified its significance in the revised manuscript. The following text has been included in the result section (Line no 223)

“The score of 1e was obtained using RBP-Var, a bioinformatics tool that scores variants involved in posttranscriptional interaction and regulation (Mao et al., 2016). Here, the annotation system rates the functional confidence of variants from category 1 to 6. While Category 1 is the most significant category and includes variants that are known to be expression quantitative trait loci (eQTLs), likely affecting RBP binding site, RNA secondary structure and expression, category 6 is assigned to minimal possibility to affect RBP binding. Additionally, subcategories provide further annotation ranging from the most informational variants (a) to the least informational variant (e). Reported 1e denotes that the variant has a motif for RBP binding. Although the employed scoring system is hierarchical from 1a to 1e, with decreasing confidence in the variant’s function. However, all the variants in category 1 are considered potentially functional to some degree.”

> *Comment 5: In Figure 3A, why is the rare SNP rs181021073 shown? This SNP does not comeup anywhere else in the paper. For clarity, it should be removed from Figure 3A.*

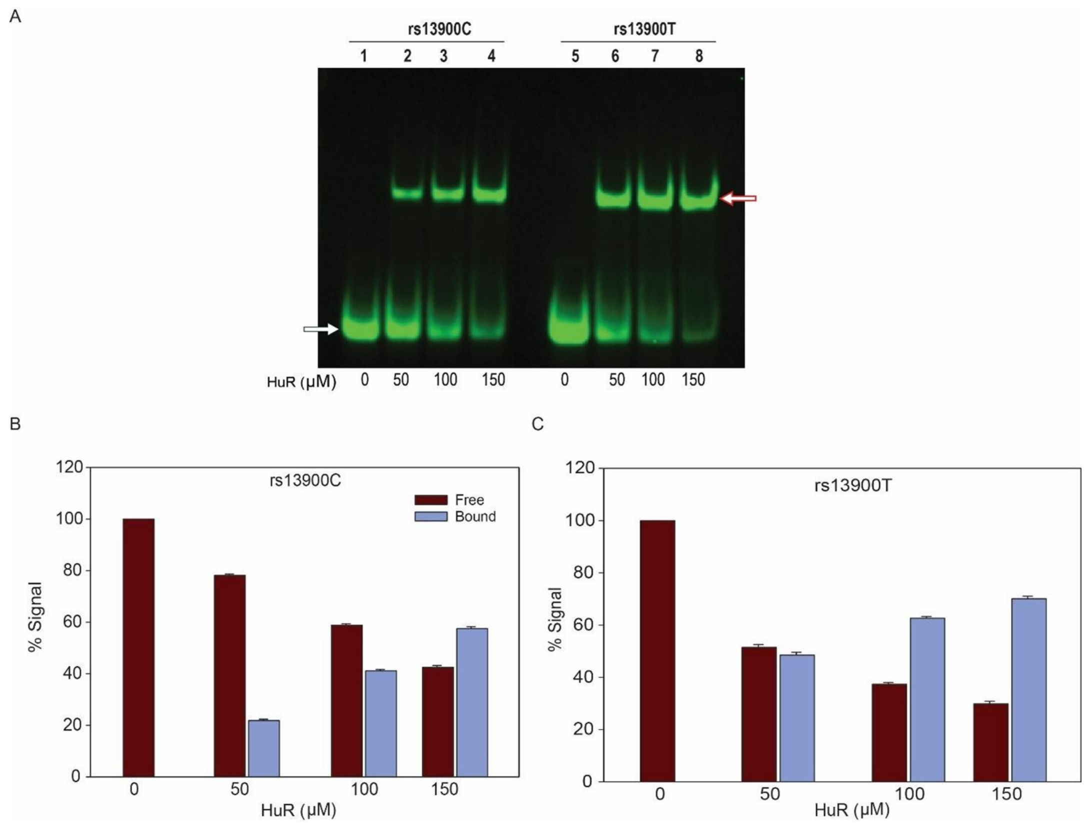
We thank the reviewer for pointing out the error in Figure 3A and apologize for the oversight. We agree that the SNP rs1810210732 is not mentioned anywhere in the manuscript and its inclusion in Figure 3A may have caused confusion. We have removed this SNP from the revised figure.

> *Comment 6: For the RNA EMSA results presented in Fig. 4C with recombinant ELAVL1 (HuR), there is clearly a loss of unbound T allele probe with increasing concentrations of the recombinant protein (without a concomitant increase in shifted complex). This suggests that the T allele probe is degraded or loses its fluorescent tag in the presence of recombinant HuR, whereas the C allele probe does not. The quantitation of the shifted complex presented in Fig. 4D as a percentage of bound and unbound probe is therefore artificially elevated for the T allele compared to the C allele. In fact, there seems to be little difference between the shifted complexes with the T and C allele probes. The authors should explain this difference in free probe levels.*

We appreciate the constructive critique of the reviewer regarding the RNA EMSA results in Fig. 4C. To address this, we repeated the experiments to analyze the differential binding of rs13900T/C allele bearing probes with increasing concentration of the recombinant HuR. No degradation/ loss of fluorescence tag for T allele was noted in presence of recombinant HuR in three independent experiments (Author response image 1). This indicates that both the probes with C or T allele show comparable stability and are not affected by increasing concentration of recombinant HuR. The apparent reduction in the unbound T allele probe in Figure 4C may be due to saturation at higher HuR concentration rather than degradation.

Author response image 1.

Differential binding and stability of oligoribonucleotide probes containing rs13900C or T alleles with recombinant HuR. (A) REMSA with labeled oligoribonucleotides containing either rs13900C or rs13900T and recombinant HuR at indicated concentrations. (B&C) Representative quantitative densitometric analysis of HuR binding to the oligoribonucleotides bearing rs13900 T or C. The signal in the bound fractions were normalized with the free probe. The figure represents data from three independent experiments (mean ± SEM).



> *Comment 7: In the Methods section, concentrations and source of reagents should be given. For example, what was the bacterial origin of LPS and concentration? What concentration of actinomycin D? What was the source? Was it provided with the nascent RNA kit? In describing the riboprobes used for REMSA, please underline the allele in the sequences (lines 549 and 550).*

Thank you for your detailed feedback and suggestions regarding the Materials and Methods Section. We regret the oversight in providing detailed information on reagent concentrations and sources in the method section. We have now rectified this omission and have provided the necessary details and a summary of material/reagents used is presented as a supplementary table (Supplementary Table 4) to enable others to replicate our experiments accurately. Regarding the description of riboprobes for RNA Electrophoretic Mobility Shift Assay, we underlined and bold the allele in the sequences as suggested (Lines 603-604).

> *Comment 8: For polysome profiling on line 603, please provide a protocol for the differentiation of primary macrophages from monocytes (please cite an original protocol, not a prior paper that does not give a detailed protocol).*

We agree with the reviewer's comment and have included the following text for primary macrophage differentiation from monocytes in the method section cited the original protocol (Line 668).

"Human monocytes were isolated from fresh blood as described earlier (Gavrilin et al., 2009) with slight modification. Briefly, peripheral blood mononuclear cells were isolated by density gradient centrifugation using Histopaque, followed by immunomagnetic negative selection using EasySep Human Monocyte isolation kit. A high purity level for CD14+ cells was consistently achieved (≥90%) through this procedure, as confirmed by flowcytometry. The purified monocytes were immediately used for macrophage differentiation by treating them with 50 ng/mL M-CSF (PeproTech) for 72 h and flow cytometric measurement of surface markers CD64+,

CD206+, CD44 was used to confirm the differentiation”. This data is now shown in the new Supplementary Figure S6.

> *Comment 9: In the legend of Figure 2, please replace "5 ug of actinomycin D" with the actual concentration used.*

We appreciate your attention to detail and thank you for pointing out the error in the legend of Figure 2. We regret the oversight and have made the suggested change (Line 739).

> *Comment 10: In the Discussion, the authors cite the study of CCL2 mRNA stabilization by HuR in mice by Sasaki et al (lines 407-9). Is regulation of CCL2 mRNA by HuR in the mouse relevant to human studies?*
>
> *How conserved is the 3'UTR of mouse and human CCL2? Is the rs13900 variant located in a conserved region? How many putative HuR sites are found in the 3'UTR of human and mouse CCL2 3'UTR? Does HuR dimerize (see Pabis et al 2019, NAR)? This information could be added to the Discussion.*

Thank you for your valuable comment. We appreciate your suggestion to include information on the dimerization of HuR in our discussion. While reporting the overall structure and domain arrangement of HuR, Pabis et al. (2019) deciphered dimerization involving Trp261 in RRM3 as key requirement for functional activity of HuR in vitro. This finding provides additional context for understanding HuR’s role in regulating *CCL2* expression. We have added the following few lines in the discussion (Lines 421-428) acknowledging HuR’s ability to dimerize and cite the relevant references.

“HuR consists of three RNA recognition motifs (RRMs) that are highly conserved and canonical in nature (Ripin et al., 2019). In absence of RNA the three RRMs are flexibly linked but upon RNA binding they transition to a more compact arrangement. Mutational analysis revealed that HuR function is inseparably linked to RRM3 dimerization and RNA binding. Dimerization enables recognition of tandem AREs by dimeric HuR (Pabis et al., 2019) and explains how this protein family can regulate numerous targets found in pre-mRNAs, mature mRNAs, miRNAs and long noncoding RNAs.”

We aligned the *CCL2* 3’UTR from five different mammalian species and found that the region flanking rs13900/ HuR binding site is relatively conserved (Author response image 2). Based on PAR-CLIP datasets there are four HuR binding regions in human *CCL2* 3’ UTR (Lebedeva et al., 2011). However, the region overlapping rs13900 seems to be predominantly involved in the *CCL2* regulation (Fan et al., 2011). This information has been included in the discussion.

Author response image 2.

Cross-species alignment of the *CCL2* 3’UTR region flanking the rs13900 using homologous regions from 5 different mammals. (Hu, Human; CH, Chimps; MO, Mouse; RA, Rat; DO, Dog, rs13900 is shown within the brackets Y, pyrimidine)

```
HU:     UU GAA UUU [Y] UGUU UAG

CH:     UU GAA UUU [U] UGUU UAG

MO:     UU GGA UUU [C] UGUU UAA

RA:     UU GGA UUU [C] UGUU UAA

DO:     UU GAA UUU [U] UGUU UAG
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> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Comment 1: The supplemental figures need appropriate figure legends.*

We regret the oversight and thank the reviewer for bringing it to our attention. We have now included the figure legend for the supplemental figures in the revised manuscript.

> *Comment 2: The data on LPS-induced CCL2 expression in PBMCs should be represented as a scatter plot with statistical significance to enhance clarity and interpretability.*

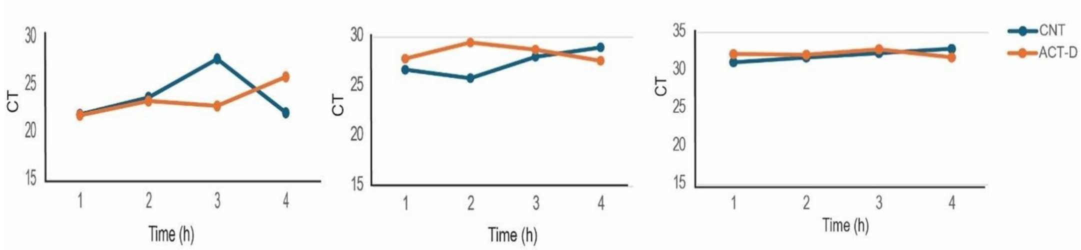
We thank the reviewer for this constructive suggestion. In the revised Figure 2A the induction of *CCL2* expression by LPS in PBMCs obtained from 6 volunteers is represented as a scatter plot. We have also included individual data points in the updated figure and statistical significance to improve clarity and interpretability.

> *Comment 3: The stability of CCL2 mRNA in control cells needs comparison with treated cells for context. The stability of a housekeeping gene (such as GAPDH or ACTB) should always be included as a control in actinomycin D experiments. Clarify the differential stability of rs13900C vs. rs13900T alleles.*

We used 18S to normalize data for the mRNA stability studies, as it is abundant and has been recommended for such studies, as it is relatively unaltered when compared to other housekeeping genes following Act D treatment in well-controlled studies (Barta et al., 2023). We also compared Ct values between the Act D-treated samples and the Act D-untreated samples in this study and found them to be comparable (Author response image 3).

Author response image 3.

Ct values of 18s rRNA in ACT-D and control samples in Fig 2.



> *Comment 4: In the main text and the methods, the authors state that nascent RNA was obtained in the presence of actinomycin D and EU. However, actinomycin D blocks the transcription of nascent RNAs, therefore the findings in Figure 2C do not reflect nascent RNA*

Please see our response to Reviewer 1 Comment 2. We would like to emphasize that to assess the differential role of the rs13900 in nascent RNA decay we integrated nascent RNA labeling and transcriptional inhibition. Briefly, PBMC from a heterozygous individual were either unstimulated or stimulated with LPS and pulsed with *5-ethynyl uridine* (0.2 mM) for 3 h and the media was replaced with EU free growth medium. RNA was obtained at 0,1, 2 and 4 h following actinomycin-D treatment (5 µg/mL) to assess the stability of nascent RNA.

> *Comment 5: Figure 4A is not clearly described or labeled. What are lanes 2 and 6?*

Figure 4 has now been updated to clearly describe all the lanes. Lanes 2 and 6 represent the mobility shift seen following the incubation by whole cell extracts and oligonucleotide bearing rs13900C and rs13900T probes respectively.

> *Comment 6: Figure 4C and Figure 4D: the charts in Figure 4D do not seem to reflect the changes in Figure 4C. How was the mean variant calculated? How do the authors explain the different quantities in unbound/free RNA in rs13900C compared to rs13900T?*

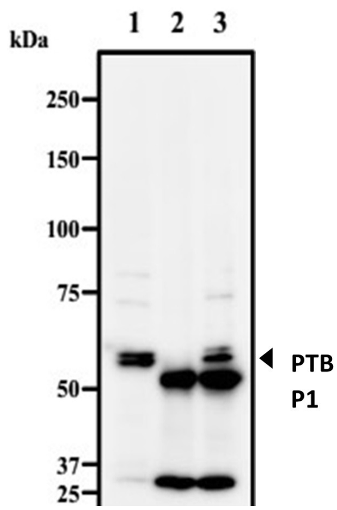
We appreciate the constructive critique of the reviewer regarding the RNA EMSA results in Fig. 4C. To address this, we repeated the experiments to analyze the differential binding of rs13900T/C probes with increasing concentration of the recombinant HuR. No degradation/ loss of fluorescence tag in presence of HuR was noted in case of T allele (Author response image 1). This indicates that both the C and T allele probes exhibit comparable stability and are not affected by increasing the concentration of recombinant HuR. The apparent reduction in the unbound T allele probe in Figure 4C may be due to saturation due to higher HuR concentration rather than degradation. Also please note under limiting HuR concentration (50µM) there is more binding of purified HuR by the T bearing oligoribonucleotide (compare lanes 2 & 6 in Author response image 1).

> *Comment 7: Figure 5A does not look like an IP. The authors should show the heavy and light chains and clarify why there is co-precipitation of beta-actin with IgG and HuR. Also, they should include input samples. Figure 5B: given that in a traditional RIP the mRNA is not cross-linked and fragmented, any region of CCL2 mRNA would be amplified, not just the 3'UTR. In other words, Figure 5B can be valuable to show the enrichment of CCL2 mRNA in general, but not the enrichment of a specific region.*

We understand the reviewer's concern on Figure 5A and 5B. Due to sample limitations we are unable to confirm these results using heavy and light chains antibodies. However, it is important to note that co-precipitation of β-actin with IgG and HuR can be due to its non-specific binding with protein G. In a recent study non-specific precipitation by protein G or A was reported for proteins such as p53, p65 and β-actin (Zeng et al., 2022). We are including a figure provided by MBL Life Sciences as the quality check document for their RIP Assay Kit (RN 1001) that was used in our study. It is evident from Author response image 4 that even pre-clearing the lysate may not remove the ubiquitously expressed proteins such as β-actin or GAPDH and they will persist as contaminants in pull-down samples. Hence the presence of β-actin in the IgG and HuR IP fractions may be due to non-specific interactions with the agarose beads.

Author response image 4.

MBL RIP-Assay Kit's Quality Check. Quality check of immunoprecipitated endogenous PTBP1 expressed in Jurkat cells. Lane 1: Jurkat (WB positive cells), Lane 2: Jurkat + normal Rabbit IgG, Lane 3: Jurkat+ anti-PTBP1.



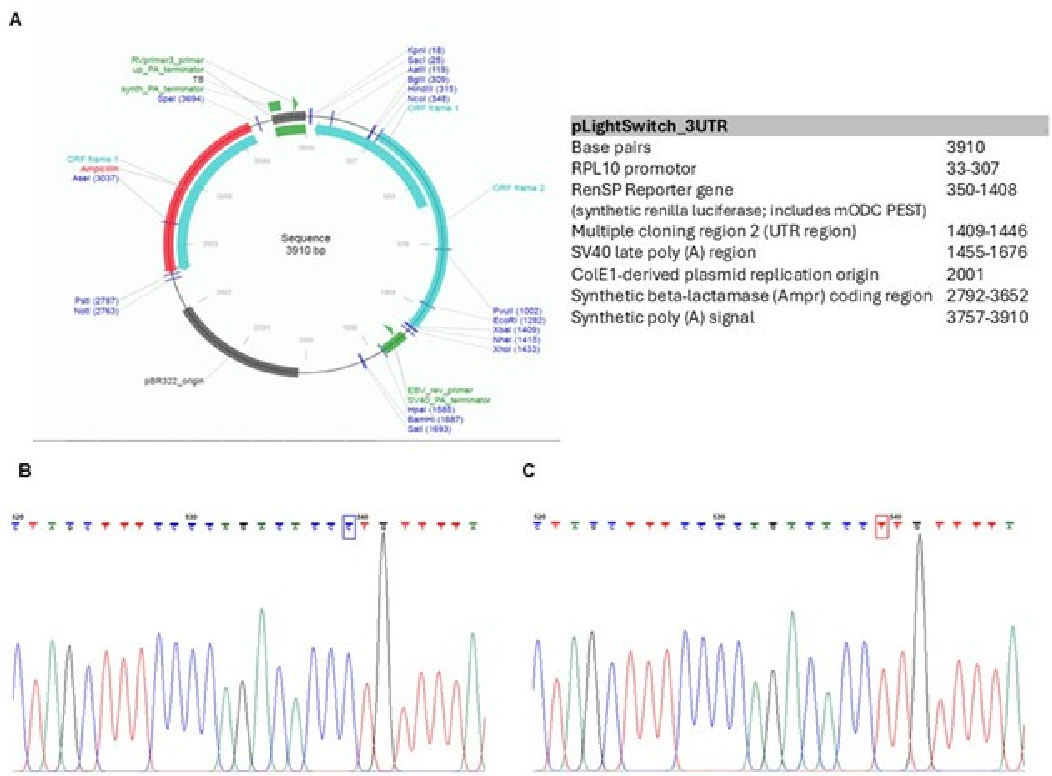
We agree with the reviewer's comments that traditional RIP without cross-linking and fragmentation allows amplification of any region of *CCL2* mRNA. However, the upregulation of *CCL2* gene expression in α-HuR immunoprecipitated samples indirectly reflects the enrichment of *CCL2* mRNA associated with HuR. Moreover, 3'-UTR targeting primers were used for amplification to examine HuR binding at this region. We believe this approach ensures that the above enrichment specifically reflects HuR association with the 3'-UTR rather than other parts of the transcript.

> *Comment 8: Construct Validation in Luciferase Assays (Figure 6): The authors need to confirm equal transfection amounts of constructs and show changes in luciferase mRNA levels. It would be better to use a dual luciferase construct for internal normalization.*

We would like to thank the reviewer for his concern and comments related to the luciferase reporter assay. As mentioned in the Methods equal transfection amount (0.5 µg) were used in our study (Line 658). We chose to normalize the reporter activity using total protein concentration instead of using a dual-reporter system to avoid crosstalk with co-transfected control plasmids. This is now included in the Materials and Method section (Lines 662-664). The optimized design of the LightSwitch Assay system used in our study allows a single assay design when a highly efficient transfection system is used (as recommended by the manufacturer). We verified the presence of the correct insert in the *CCL2* Light Switch 3'UTR reporter constructs (Author response image 5). We also sequenced the vector backbone of both constructs to rule out any inadvertently added mutations.

Author response image 5.

Schematic of the Lightswitch 3'UTR vector. (A) Vector information. The vector contains a multiple cloning site (MCS) upstream of the Renilla Luciferase gene (RenSP). Human 3'UTR *CCL2* is cloned into MCS downstream of the reporter gene and it becomes a part of a hybrid transcript that contains the luciferase coding sequence used to the UTR sequence of *CCL2*. Constructs containing rs13900C or rs13900T allele were generated using site-specific mutagenesis on *CCL2* LightSwitch 3'UTR reporter. The constructs were validated by Sanger sequencing. (B&C) Sequence chromatograph of the constructs containing *CCL2*-3'UTR insert showing rs13900C and rs13900T respectively. The result confirms the fidelity of the constructs used in the reporter assay.

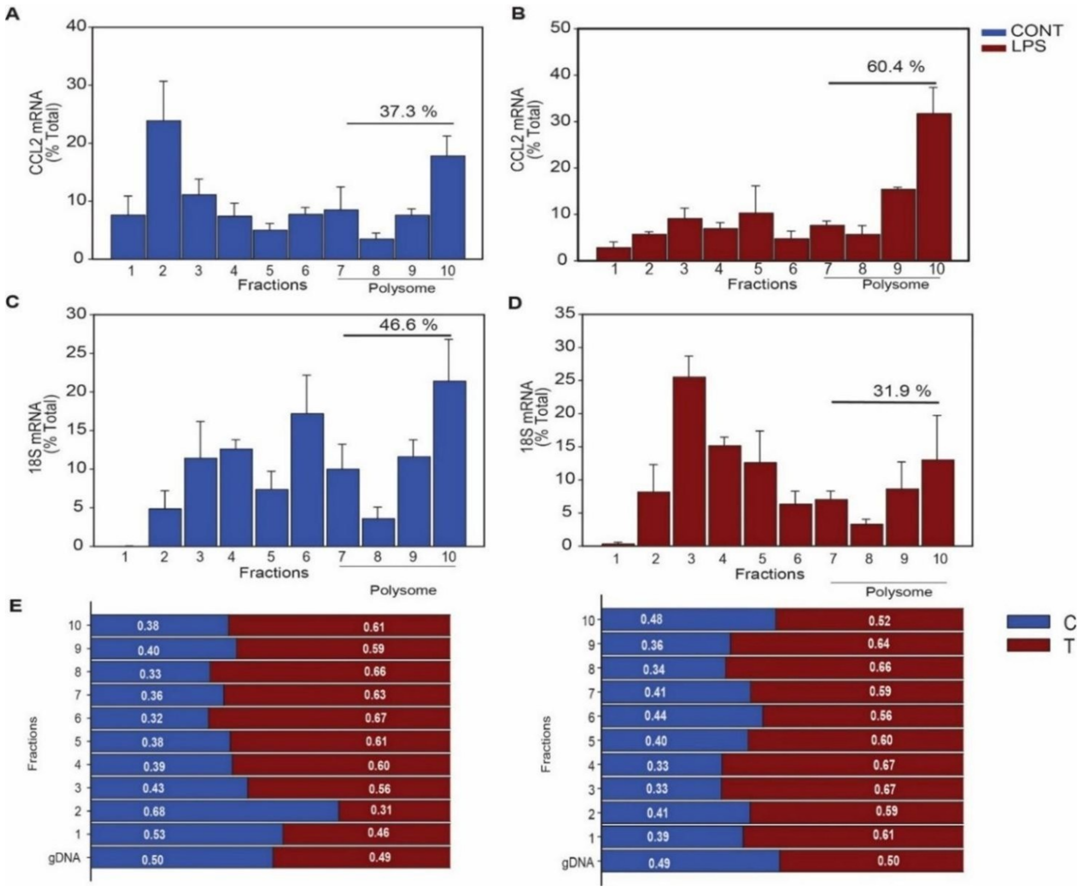


> *Comment 9: Polysome Data Presentation: The authors should present the distribution of luciferase mRNA (rs13900T and rs13900C) in all fractions separately and include data on the translation of a control like ACTB or GAPDH.*

Since our assessment of *CCL2* allele-specific enrichment in the polysome fractions from MDMs of heterozygous donors did not yield a consistent pattern for differential loading (Supplementary Table3), we used a 3'UTR reporter-based assays that estimated the impact of rs13900 T and C alleles on overall translational output (translatability). The translatability was calculated as luciferase activity normalized by luciferase mRNA levels after adjusting for protein and 18S rRNA using a previously reported method (Zhang et al., 2017). As the measurement of relative allele enrichment in polysome fractions was not included in our invitro reporter assays, it is not possible to present the distribution of luciferase mRNA in various fractions separately. Author response image 6 shows the proportion of *CCL2* mRNA in different fractions corresponding to cytosolic, monosome and polysome fractions obtained from MDM lysates from heterozygous donors along with 18S rRNA quantification.

Author response image 6.

Determination of rs13900C/T allelic enrichment in polysome fractions and its effect on polysome loading. Polysome profile obtained by sucrose gradient centrifugation of macrophages before and after stimulation with LPS (1 µg/mL) for 3 h. (A&B) The *CCL2* mRNA shifts from monosome-associated fractions to heavier polysomes following LPS stimulation, indicating increased translation efficiency. (C&D) In contrast, the distribution of 18S shows no significant shift due to LPS treatment. (mean ± SEM, n=4). The percentage of mRNA loading on polysome was calculated using ΔCT method (mean ± SEM, n=4). (E&F) *CCL2* AEI measurement in polysomes of macrophages from heterozygous donors (n=2). Genomic and cDNA were subjected to Sanger sequencing and the peak height of both the alleles were used to determine the relative abundance of each allele.



> *Comment 10: Please explain in detail how primary monocytes were transfected with siRNAs for more than 72 hours. Typically, primary monocytes are very hard to transfect, have a very limited lifespan in culture (around 48 hours), and show a high level of cell death upon transfection. If monocytes were differentiated from macrophages, explain in detail how it was done and provide supporting citations from the literature.*

We agree with the challenges associated with transfecting primary monocytes, including their limited lifespan in culture and susceptibility to cell death following transfection and apologize for not elaborating the method section on lentiviral transduction of primary macrophages. To overcome these limitations, we utilized monocytes undergoing differentiation into macrophages rather than fully differentiated macrophages for our experiments. Cells were transfected by slightly modifying the method described by Plaisance-Bonstaff et.al 2019 (Plaisance-Bonstaff et al., 2019). Briefly, monocytes were purified from PBMCs obtained from homozygous donors for rs13900 C or rs13900T by negative selection. Upon purification cells were resuspended in 24 well plates at a seeding density of 0.5 x10^6 cells per well and were further cultured in the medium supplemented with 50 ng/mL M-CSF (Fig S7 and Fig. S6). After 24 h, ready to use GFP-tagged pCMV6-HuR or CMV-null lentiviral particles (Amsbio, Cambridge, M.A) were transduced into 0.5 x10^6 cells in presence of polybrene (60 µg/mL) at a MOI of 1. The cells were processed for HuR and *CCL2* expression 72 h after transduction after stimulation with LPS for 3 h. This data is now shown in new Supplementary Figure S7.

> *Comment 11: The authors should prove the binding of HuR to the 3'UTR of CCL2 not only in vitro but also in cells. For this aim, a CLIP including RNA fragmentation followed by RT-PCR or sequencing would be more informative than a RIP. It would be helpful also to demonstrate the different binding to the 3'UTR variants (rs13900C vs. rs13900T).*

We thank the reviewer for his valuable suggestion on validating binding of HuR to the 3'UTR in cells. It is important to highlight that several independent datasets including CLIP have already demonstrated that HuR binds to the 3'UTR of *CCL2* including the region spanning the rs13900 locus. We have summarized the relevant studies in a tabular form (Supplementary Table-2). We are unable to confirm these results in new experiments due to sample limitation. The already existing data and experimental evidence provided in this manuscript strongly suggest that HuR binds within the 3'UTR. Also, a previously published study (Fan et al, 2011) showed that only the first 125 bp of the *CCL2* 3'UTR that flanks rs13900 showed strong binding to HuR but not the CCL2 coding region or other regions of 3'UTR. This further suggests that the HuR binding to the *CCL2* is localized to the 3'UTR that flanks rs13900. Please note that the primers used for amplification of the RIP material were 3'-UTR specific.

> *Comment 12: To quantify nascent RNA, Figure 2C should be replaced by new experiments. To label nascent RNA, authors can perform a run on/run-off experiments only with EU, without actinomycin D. As aforementioned, ActD blocks the transcription of new RNA, therefore is not useful for studying nascent RNA.*

We thank the reviewer for the suggestion and would like to emphasize that while measuring the rs13900C/T allelic ratio in nascent RNA, the experimental setup included evaluating the AEI both in presence and absence of the transcriptional inhibitor actinomycin D. The data presented in Figure 2C shows that the AEI in presence of actinomycin D is amplified in comparison to non-actinomycin D treatment. This provides definitive evidence to our hypothesis that rs13900T confers greater stability to the *CCL2* message. We apologize for the oversight of not mentioning non-ACT D treatment in the methods. Necessary changes have been made to the revised manuscript (Lines 553-63).

> *Comment 13: The authors should also investigate the role of TIA1 as a potential RBP and explore the possibility that TIA1 may interact more with the C allele to suppress translation.*

Based on the existing studies, we highlighted the importance of RNA-binding proteins such as TIA1 and U2AF56 that may interact with *CCL2* transcript (Lines 408-09). However, exploring TIA1 binding and its functional consequences are beyond the scope of the current study. We thank the reviewer for this comment and this aspect will be pursued in future studies.

> *Comment 14: It would be informative if the authors included study limitations and potential clinical implications of these findings, particularly regarding therapeutic approaches targeting CCL2.*

We would like to inform the reviewer that the submitted manuscript included the limitations of our study. They were discussed at appropriate places and were not included as a separate section. For instance, Line 398 emphasizes the need for in-depth studies for association of rs13900 and canonical *CCL2* transcript. The need for additional studies regarding SNP-induced structural changes in RNA and its implication for RBP accessibility was highlighted at Lines 417-419. The inconclusive results of differential loading of polysomes and the need to conduct further research on the impact of rs13900 on CCL2 translatability in primary cells (Lines 457-459). We noted at Lines 484-485 about our further studies exploring the differential binding of HuR to the other regions of *CCL2* 3'UTR.

Multiple studies have indicated that functional interference of HuR as a novel therapeutic strategy, particularly in the context of cancer, inflammation, neurodegeneration, and autoimmune disorders. These approaches include inhibitors such as MS-444, KH-3, and CMLD-2 that disrupt the interaction between HuR and ARE elements or mRNAs of target genes involved in disease pathology (Chaudhary et al., 2023; Fattahi et al., 2022; Lang et al., 2017; Liu et al., 2020; Wang et al., 2019; Wei et al., 2024), offering a potential new avenue for disease treatment. Findings from our studies provide unique insights on regulation of CCL2 expression by both rs13900 and HuR. We strongly believe that the SNP rs13900 and HuR represent a new druggable target for M/M-mediated disorders such as inflammatory diseases, cancer, and cardiovascular diseases. The potential clinical implications have been discussed in the revised manuscript (Lines 487-494)

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